Do Small Shareholders Have a Voice?
Deliberation, Delegation, and Value in DAO Governance*

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Abstract

We study whether small token holders have voice in the governance of decentralized autonomous organizations (DAOs), and whether that voice creates value. We develop a model in which the informed party is the protocol user rather than the large holder: users learn the long-run consequences of governance through protocol usage. Discussion forums aggregate users' dispersed information, and delegation concentrates their dispersed votes into a decisive bloc and lowers the cost of participation. Using account-level data on 2,830 proposals from DeFi DAOs, we find that these features raise small holders' participation and influence over outcomes. Small-holder victories are rare, but when they occur, small holders prevail by pooling their votes through delegation rather than by allying with large holders. Their victories earn positive abnormal returns, and the reaction is larger when the preceding deliberation reached more consensus. The prevailing holders hold their tokens rather than sell.

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1 Introduction

Decentralized autonomous organizations (DAOs) emerged with the promise of broad and inclusive governance. Any token holder can submit a proposal, join the deliberation, and vote, without the intermediaries that limit participation in corporate governance. In practice, however, voting power is highly concentrated, and a small number of large holders determine most outcomes under the one-token-one-vote standard (Fritsch et al., 2022; Han et al., 2026). This gap between the inclusive ideal and the concentrated reality motivates our question, which is familiar from corporate governance: do small shareholders have any voice, and does it matter for value?[1] In this paper, we use the institutional setting specific to DAO governance to study whether and how the voice of small token holders can be aggregated and reflected in governance outcomes, and whether it improves the value of the protocols they govern.

The corporate governance literature offers a clear benchmark. Dispersed small shareholders are thought to be uninformed and rationally apathetic, because each holds too little to justify the cost of becoming informed and voting, while large blockholders are the informed monitors who discipline management (Shleifer and Vishny, 1986; Grossman and Hart, 1980; Maug, 1998). The DAO setting inverts part of this benchmark. The dispersed small token holders who hold a governance token are often the active users of the protocol, who experience its quality directly through usage. This usage plays a dual role: it provides active users with more precise private signals about the long-run consequences of governance decisions, and it exposes them to more of the protocol degradation that a value-destroying proposal imposes, so heavier users have both better information and higher stakes in blocking bad proposals. The large holders, by contrast, are frequently financial investors who observe the protocol mainly through financial metrics. The party with the most relevant private information about the long-run consequences of a governance decision can therefore be the small rather than the large token holders. The friction is that these informed users are also the most dispersed, and so face the collective action problem most acutely.

What distinguishes DAO governance from corporate voting is the infrastructure that surrounds the vote. Proposals are deliberated in public discussion forums before voting opens. Token holders can delegate their voting rights to representatives at low cost. Every ballot, the

[1]Throughout, we use "shareholder" by analogy to the corporate governance literature. Holders of a DAO governance token are not shareholders in the legal sense: the token confers an ERC-20 governance right rather than an equity claim, and it differs from a corporate share in that it trades continuously and is observable at the level of the individual account. We adopt the term because our question and benchmark are drawn from that literature, and we use "token holder" as the precise term in the model and empirical analysis. "Small shareholder" and "small token holder" refer throughout to the same object, i.e., a holder of less than 5% of circulating supply.

voting power behind it, every delegation, and every token transfer is recorded on a public ledger. This infrastructure gives dispersed users tools to overcome the collective action problem, and it gives us account-level data that is not available in the study of corporate shareholder voting. We observe who participates, how they participate, what is said in the deliberation that precedes the vote, and how each participant trades after it, each at the level of the individual account. Studies of corporate votes must instead infer the value of a governance event from prices alone.

We develop a model of DAO governance to organize this analysis. Token holders differ along two dimensions: the weight they place on financial returns relative to protocol utility, and the intensity with which they use the protocol. A proposal can extract value from the protocol in a way that raises near-term token value but degrades long-run protocol health. Active users observe noisy private signals about this degradation through their usage experience—with more active users receiving more precise signals—while financial investors observe only short-run financial metrics and hold more optimistic priors about long-run protocol health. Because the long-run cost of a value-destroying proposal is borne in proportion to usage, usage intensity simultaneously sharpens each user's private information and raises her stake in blocking. Without governance infrastructure, the informed users are non-pivotal and do not participate, so financial investors determine outcomes and value-destroying proposals pass. This is the DAO analog of the free-rider problem in Grossman and Hart (1980).

Governance infrastructure relaxes these frictions through two complementary mechanisms that address two distinct problems. Discussion forums address the information and coordination problem. They aggregate dispersed private signals into a public signal that becomes more precise as more users contribute, raising each user's information about proposal quality; and because higher-quality, more consensual deliberation is easier for an outside observer to corroborate, the forum is more credible the better its quality, so users weight it more heavily than financial investors who cannot verify it against direct experience. Forums also serve a coordination role: observing broad participation in the discussion raises a user's expectation that like-minded holders will act, and hence the perceived benefit of her own participation. Delegation addresses the collective-action problem. It concentrates dispersed votes into a bloc that is pivotal with positive probability, restoring a participation incentive that an atomistic holder acting alone does not have, and it lowers the per-proposal cost of participation. Because the benefit of delegating rises with a user's own usage intensity—which determines both the stake she has in blocking and the precision of her signal—it is the heavy users who cross the participation threshold and form the bloc, so the participating coalition is composed of the better-informed users. The model delivers four predictions. Governance infrastructure raises the participation of active users, primarily through the delegation margin, with the largest effect among heavy protocol users. It raises the probability that informed users determine outcomes by shifting the composition of participants toward those with more precise signals. Outcomes in which users block value-destroying proposals are associated with positive abnormal returns. The size of this valuation effect is increasing in the quality of the deliberation, not merely its presence.

We test these predictions on a sample of 2,830 governance proposals from DAOs that govern DeFi protocols on Ethereum. We obtain proposal and ballot-level vote data from Snapshot, token prices from CoinGecko, token holdings and transfers from Ethereum archive nodes, and discussion threads from the protocols' public forums. We link each voting address to its transaction history to determine whether the holder has used the protocol it governs, and we use a large language model to measure the consensus of each discussion thread. We classify holders below five percent of supply as small token holders and the rest as whales, and we classify a proposal as a small holders' victory when the outcome matches the majority choice of small token holders but differs from that of whales.

The data confirm the inversion at the center of the model. Small token holders are far less engaged than whales along every participation margin: their turnout is 2.6%, against 52.9% for whales. Yet they are more likely than whales to have used the protocol they govern, with a usage rate of 0.68 against 0.56. The dispersed holders are thus the better-informed users, and the participation gap is the empirical counterpart of their rational apathy.

We find that governance infrastructure raises small holder participation, as the model predicts. The availability of delegation, the presence of discussion, the consensus reached in discussion, and confirmed protocol usage all raise participation, while weighted voting, which concentrates voting power, lowers it. Small token holders participate more in proposals that bear directly on the protocol they use, such as proposals on protocol security and user incentives, which is the signature of the usage-based information channel. To address the concern that DAOs adopting delegation differ from those that do not, we exploit the staggered adoption of delegation in a difference-in-differences design. Enabling delegation raises small token holder participation by 1.5 percentage points, an effect that emerges only after adoption and is large relative to the sample mean of 3.2%.

We then show that this participation translates into influence and value. Small holder victories, outcomes that match the small-holder majority but not the whale majority, are rare: they occur on only 6.4% of proposals, and on most proposals concentrated holders prevail. We do not claim that small token holders routinely win. Our interest is in the opposite question that this rarity makes pressing: when they do win, how do they do it, and what makes it possible? The same features that raise participation, delegation, discussion consensus, and confirmed usage, also raise the probability of a small-holder victory. Small-holder victories are in turn associated with a cumulative abnormal return of about 4.5% around the proposal, against a general negative reaction for proposals on average. The market treats a small-holder victory as good news, consistent with the model's prediction that such an outcome reveals that a large value destruction was avoided. Consistent with the model's further prediction that this valuation effect rises in the quality of deliberation, around voting end the abnormal return to a small-holder victory is larger when the discussion preceding the vote reached more consensus.

We use the account-level data to examine how small holders win, and find that they prevail by pooling their own votes rather than by allying with sympathetic large holders. Almost all victories occur in DAOs that enable delegation, and among these, the winning side draws roughly three quarters of its participation from delegated rather than directly cast votes. Reconstructing the voting power on each side of a victory, small token holders supply about 97% of the winning side, and in most victories no whale appears on the winning side at all. Whales, large and small, are predominantly on the losing side. Small holders therefore overcome the collective action problem by aggregating their dispersed votes through delegation, not by realigning the largest holders.

Finally, we use the account-level transfer data to examine how small holders trade after a vote. Small holders who prevail sell less, not more, in the days following the outcome. They hold their tokens rather than realize the positive price reaction, which is consistent with the premise that the participating small holders are protocol users whose holdings reflect long-run alignment with the protocol rather than short-run financial speculation.

Taken together, the theory and the evidence show that the institutional features specific to DAO governance improve the position of small token holders. The deliberation forum and delegation help dispersed but informed users overcome the coordination and collective action problem, raising their participation and their influence over outcomes. As the better-informed users come to determine more outcomes, governance becomes more informative, and the market rewards the outcomes that block value-destroying proposals. Governance infrastructure therefore makes DAO governance both more inclusive and more value-creating.

These lessons extend beyond decentralized governance. The two mechanisms at the center of our model—deliberation and delegation—each have counterparts in traditional corporate governance whose design our evidence speaks to directly. Traditional corporate governance lacks a structured, transparent channel through which dispersed retail token holders can aggregate their private information before a vote. Proxy advisory firms fill part of this role, but from the institutional rather than the retail side: they aggregate information on behalf of large asset managers, not from the dispersed users who may have superior knowledge of the firm's products and operations. Our evidence suggests that what matters is not merely the presence of deliberation but its credibility and convergence. Proposals preceded by more consensus-driven discussion attract higher small token holder participation and earn larger market rewards when small token holders prevail, pointing to a quality channel distinct from the breadth of coverage that proxy advisors provide. This finding motivates structured, publicly observable pre-vote deliberation platforms in corporate settings as a complement to existing proxy processes—a mechanism that could help dispersed, operationally informed retail token holders pool their knowledge, increase the perceived value of participating, and arrive at the ballot better prepared to exercise collective voice.

The delegation channel points to a related but distinct policy conclusion. In traditional corporate voting, retail investors routinely delegate their votes to institutional investors—mutual funds and index funds—that vote on their behalf, and this delegation has historically been the primary channel through which small shareholders exercise collective voice. A recent and growing policy debate sharpens this channel. Major fund managers including BlackRock and Vanguard have introduced voting-choice programs that offer their investors the option to vote their own shares rather than delegate to the fund—so-called pass-through voting. Malenko and Malenko (2024) show that such programs improve investor welfare when investors have heterogeneous information about the proposal, precisely the informational environment our model identifies: dispersed users who know more than financial intermediaries about the operational consequences of a governance decision. Our findings reinforce their conclusion with direct empirical evidence. In the DAO setting, where small users are the informed party, delegation that aggregates their private signals into a coordinated voting bloc improves governance quality and protocol value. The implication for traditional finance is that both channels matter—credible deliberation that surfaces retail token holders' dispersed operational knowledge, and delegation structures that concentrate their coordinated voice into a decisive vote—and that designing governance around the informed party, rather than the largest one, can make corporate decisions both more inclusive and more value-creating.

The remainder of the paper is organized as follows. Section 2 reviews the related literature, followed by an introduction of institutional background of DAO governance in Section 3. Section 4 presents the theoretical framework and its predictions. Section 5 describes the data and sample, and Section 6 sets out the empirical specifications. Section 7 reports the results. Section 8 concludes. The appendix collects the proofs and additional details.

2 Related Literature

Our paper relates to two strands of literature: the literature on shareholder voting and the role of small shareholders in corporate governance, and the literature on the governance of decentralized organizations.

2.1 Shareholder voting and the role of small shareholders

A large literature studies how shareholder rights and voting affect firm value. Gompers et al. (2003) develop a shareholder rights index and document that stronger shareholder rights are associated with higher firm value and profitability, and Bebchuk et al. (2009) construct a similar index to show that voting mechanisms adverse to shareholders, such as staggered boards, are associated with negative abnormal returns. While these index-based studies establish a correlation between governance and value, a subsequent literature isolates the causal effect of the vote itself. Using a regression discontinuity design around the majority threshold, Cuñat et al. (2012) show that passing a shareholder-sponsored governance proposal raises firm value by roughly 2.8%. Yermack (2010) surveys the growing evidence that the mechanics of shareholder voting shape governance outcomes. Together these findings establish that not only the design but also the exercise of shareholder voting matters for value.

A central question in corporate governance literature is which shareholders are informed and who monitors. The classical view holds that large blockholders are the informed monitors who discipline management, while dispersed small shareholders are rationally apathetic because each is too small to justify the cost of becoming informed and voting (Shleifer and Vishny, 1986; Grossman and Hart, 1980; Maug, 1998). Recent evidence on retail investors qualifies this view: Brav et al. (2022) use a near-comprehensive sample of U.S. ownership and voting records to show that, conditional on participating, small shareholders monitor their portfolio firms and discipline the management of poor performers, rather than deferring mechanically to management. A related literature studies how shareholders coordinate and how trading interacts with voting. Levit and Malenko (2011) and Malenko and Malenko (2019) model the coordination of dispersed shareholders, and Levit et al. (2024) shows that the ability to trade does not necessarily produce optimal voting outcomes, because shareholders with extreme positions can amass stakes and use their voting power at the expense of moderate shareholders and overall welfare. Closest to our question of who should hold the vote, Malenko and Malenko (2024) show that when investors are heterogeneous in information, placing voting power with the individual investors who hold the relevant private signals improves welfare. We contribute to this literature by studying a setting in which the informational roles are reversed. In a DAO the dispersed small holders are often the active users of the protocol, who hold the relevant private information about the long-run consequences of a governance decision, while the large holders are financial investors who observe the protocol mainly through financial metrics. Information heterogeneity, rather than preference heterogeneity, is therefore the first-order force, so that—consistent with Malenko and Malenko (2024)—placing voting power with informed users is desirable. Our account-level data also let us measure the quality of participation, and not merely its quantity, through the deliberation that precedes each vote.

2.2 Governance of decentralized organizations

A growing literature studies governance on public blockchains. Yermack (2017) assesses the implications of blockchain technology for corporate governance and emphasizes lower costs, more reliable record-keeping, and greater transparency. Fritsch et al. (2022) document the concentration of voting power in the DeFi protocols Compound and Uniswap, where it is held by a small number of Ethereum addresses. Han et al. (2026) model the dynamics of DAO governance and the interaction between ownership concentration and strategic token trading. They find a conflict of interest between major token holders, who may pursue self-serving proposals, and minority holders, and they show that token concentration harms governance and the health of the DAO. In their model, open-market trading of governance tokens lets dominant actors accumulate holdings and sway voting outcomes for short-term benefits, to the detriment of minority stakeholders. Closer to our valuation results, Appel and Grennan (2023a,b) find that DAO proposals are typically highly centralized, but that DAOs adopting inclusive voting mechanisms that engage a broad set of token holders and promote security earn positive excess returns on their tokens. Other recent work studies the organization and decision-making of DAOs more broadly (Fracassi et al., 2024; Sun et al., 2024; Laturnus, 2023).

We complement this literature in two ways. First, we provide a theoretical foundation in which the informed party is the protocol user rather than the large holder, which inverts the standard assumption of the corporate governance literature (Shleifer and Vishny, 1986; Maug, 1998). Second, we exploit account-level data on votes, delegation, deliberation, and trading that is not available in studies of corporate voting. This lets us trace the channel from deliberation to participation, to influence over outcomes, and to value, and to show how the governance infrastructure specific to DAOs helps dispersed but informed users overcome the collective action problem.

3 Institutional Background

DAOs operate on public blockchains, most commonly Ethereum. Governance rights are represented by an ERC-20 token (Vogelsteller and Buterin, 2015), a fungible token standard that records balances and transfers on the blockchain. A governance token confers the right to vote on proposals, with voting power proportional to the number of tokens held under the one-token-one-vote standard. Governance tokens differ from corporate shares in two ways that matter for our analysis. First, they trade continuously on centralized and decentralized exchanges, so voting power can be acquired or sold at any time, whereas a shareholder who wishes to vote without holding shares must borrow them. Second, every token holding, transfer, and vote is recorded on a public ledger and is observable at the level of the individual account.

A DAO governs a protocol through a two-stage process. A proposal is typically preceded by an off-chain discussion thread, followed by proposal creation, the opening and closing of voting, and on-chain implementation, as Figure 1 illustrates. This sequence gives two natural event dates, proposal creation and voting end, around which we measure trading and returns. A proposal is first deliberated in a public discussion forum, and proposals that attract sufficient support proceed to a voting stage in which token holders cast votes. Uniswap, a leading decentralized exchange, is a representative example: its community deliberates on proposals and then votes, with voting power tied to Uniswap token holdings under the one-token-one-vote principle (Adams et al., 2021). Most DAOs conduct voting off-chain through Snapshot, the leading voting infrastructure, which records votes without incurring transaction fees. Voting power is fixed at a snapshot of token balances taken at a designated block, which limits manipulation through last-minute borrowing. Holders may also delegate their voting rights to representatives, who vote on their behalf. DAOs differ in the voting mechanism they adopt, including the one-token-one-vote rule, weighted voting, and quadratic voting. Buterin et al. (2019) show that quadratic voting, under which the cost of acquiring votes rises convexly with voting power, can be superior when agents have well-defined preferences over outcomes, while Benhaim et al. (2023) show that quadratic voting can be inferior to linear or one-token-one-vote rules when agents are uncertain about proposal quality.

A potential concern with token-based voting is that it concentrates influence among the largest holders, who can more easily coordinate to pass proposals than dispersed small token holders, and that it is open to vote buying through the open-market trading of governance tokens. Proposed mitigations include delegation, which lets dispersed holders pool their voting power, and non-transferable governance tokens, which would prevent the accumulation of voting power through trading. These features of the institutional setting motivate the model and the empirical analysis that follow.

4 Theoretical Framework

We develop a reduced-form model of DAO governance that formalizes the tension between financially-oriented token holders and informed protocol users to guide our empirical investigation. The central friction is informational: active protocol users possess superior knowledge about the long-run consequences of governance decisions because they directly experience protocol quality through usage, whereas financially-oriented token holders primarily observe short-run financial metrics and therefore may support extraction policies that increase near-term token value while degrading long-run protocol health. We show that governance infrastructure, specifically discussion forums and delegation mechanisms,can help dispersed protocol users overcome collective action problems and prevent value-destroying proposals from passing. The framework generates four testable predictions that guide our empirical analysis.

4.1 Environment

Agents. A DAO governs a protocol through token-based voting. To capture the empirically relevant distinction between large token holders who individually affect governance outcomes and dispersed small holders whose individual influence is negligible, we model the economy as a mixed population following Shleifer and Vishny (1986) and Maug (1998).

Specifically, there is a finite set of large token holders (hereafter, *financial investors*) indexed by $k \in \{1, \dots, K\}$, each holding a discrete stake share $\alpha_k^W > 0$. These agents possess non-negligible voting power and internalize a meaningful fraction of the financial consequences of governance outcomes. There is also a continuum of atomistic token holders indexed by $h \in [0, 1]$, each holding an infinitesimal stake share $d\alpha_h^S$. Total token supply is normalized to one:

$$\sum_{k=1}^{K} \alpha_k^W + \int_0^1 d\alpha_h^S = 1. \tag{1}$$

Voting power is proportional to token holdings under the one-token-one-vote protocol standard in DAO governance.

Each token holder is characterized by two independent dimensions of heterogeneity:

(i) *financial orientation* $\theta_h \in [0, 1]$, capturing the relative weight placed on financial returns versus protocol utility; and

(ii) *protocol usage intensity* $\delta_h \in [0, \delta_{\max}]$, capturing the degree to which the agent actively interacts with and derives utility from the protocol.

We place no restrictions on the joint distribution of (θ_h, δ_h) beyond mild regularity conditions. Note that neither dimension is directly observable in governance data; we discuss the empirical mapping from observables to these theoretical primitives in Section 4.2 below.

Proposals. At each governance event, token holders vote on a binary proposal $a \in \{0, 1\}$, where $a = 1$ denotes an *extraction policy* and $a = 0$ denotes the *status quo*. Extraction policies redistribute value from the protocol to token holders in ways that degrade long-run protocol health; examples include fee increases that redirect protocol revenue to token holders at the expense of user adoption, treasury distributions that deplete reserves earmarked for ecosystem development, or tokenomics changes that boost near-term token prices at the cost of long-run competitive position. The status quo preserves existing protocol parameters.

Preferences. All token holders share the same objective function. Token holder h's net payoff from the extraction proposal relative to the status quo is:

$$U_h(a = 1) - U_h(a = 0) = \underbrace{\theta_h \cdot \alpha_h \cdot \mathbb{E}_h[\Delta V]}_{\text{financial effect}} - \underbrace{(1 - \theta_h) \cdot \delta_h}_{\text{usage cost}}, \tag{2}$$

where $\mathbb{E}_h[\Delta V]$ is token holder h's expectation of the net change in token value under extraction, α_h denotes stake size (either α_k^W for financial investors or $d\alpha_h^S$ for atomistic holders), and $\delta_h \geq 0$ is the capitalized protocol utility loss experienced under extraction, equal to a per-period loss summed over the holder's usage horizon. Both components of the payoff are therefore expressed in present-value terms: the financial effect is a capitalized change in token value, and δ_h is the capitalized usage loss. The financial weight θ_h determines the trade-off between these components: high θ_h implies the financial effect dominates, while low θ_h implies protocol utility is relatively more important. Note that the usage cost is weighted by $(1 - \theta_h)$, so a financially-oriented token holder who also uses the protocol effectively discounts their own usage costs.

Typology of token holders. The two-dimensional heterogeneity in (θ_h, δ_h) naturally partitions token holders into four economically distinct groups, summarized in Table 1. This typology is not imposed as a maintained assumption; it emerges from the parameter space and provides a useful organizing framework for the model's predictions.

Financial investors place high weight on financial returns and interact minimally with the protocol. Their payoff from extraction is approximately $\theta_h \cdot \alpha_h \cdot \mathbb{E}_h[\Delta V]$, and because $\delta_h \approx 0$, their usage cost is negligible. They do not directly observe the long-run consequences of governance decisions through usage, making them informationally disadvantaged with respect to the unobservable value component, as we formalize in Section 4.2. In practice, financial investors correspond to venture capital funds, protocol treasuries, early-stage investors, and quantitative funds holding governance tokens primarily for financial exposure.

Protocol users place low weight on financial returns but actively use the protocol, exposing them directly to its consequences. Their payoff is dominated by $-(1-\theta_h)\delta_h < 0$, making them strongly opposed to value-destroying extraction. Their active engagement with the protocol generates precise private signals about long-run protocol quality. In practice, protocol users correspond to retail participants who interact with the protocol frequently (liquidity providers, active traders, or frequent borrowers in DeFi protocols), and who typically hold small token positions relative to their usage intensity.

Investor-users combine high financial orientation with high usage intensity, giving them incentives on both dimensions to oppose value-destroying extraction while remaining financially sophisticated. Their dual role makes them natural delegation representatives who can translate dispersed user information into effective governance power, as we discuss in Section 4.4. In practice, investor-users correspond to protocol developers, core contributors, and founding team members who retain significant governance stakes while remaining active protocol participants. This group is illustrative rather than a formal element of the analysis: the equilibrium results below are derived from the two polar types, the financial investor and the protocol user, and investor-users serve only to interpret who acts as a delegate in practice.

Passive holders place low weight on both financial returns and protocol utility, making them largely indifferent between proposals. Their rational indifference contributes to the low overall governance participation rates documented in the prior literature.

4.2 Information Structure

The central friction in our model is that the net value effect of extraction, ΔV, is not fully observable to all token holders, and the resulting information gap is not symmetric across types.

Value decomposition. The net change in token value under extraction is:

$$\Delta V = \pi^O + \pi^U, \tag{3}$$

where $\pi^O > 0$ denotes the *observable* value component and $\pi^U < 0$ denotes the *unobservable* value component. The observable component captures publicly verifiable short-run financial effects of the extraction policy (announced fee increases, treasury distributions, or token buyback schedules), whose implications are straightforward to assess from on-chain data and protocol documentation, and which therefore constitute common knowledge. The unobservable component captures the capitalized loss from long-run protocol degradation (user attrition, reduced ecosystem investment, or competitive disadvantage relative to rival protocols), which manifests gradually and can be assessed primarily through active protocol usage. We focus throughout on the empirically relevant case:

Assumption 1 (Value destruction)**.** *$|\pi^U| > \pi^O$, so that $\Delta V = \pi^O + \pi^U < 0$. The extraction proposal is value-destroying in expectation for a fully informed token holder.*

Assumption 1 defines the class of proposals for which governance quality matters: those that appear financially attractive on observable metrics but are value-destroying once long-run protocol degradation is accounted for. The valuation result of Proposition 4 conditions on this class: it is stated for proposals the forum and the ex-post outcome identify as value-destroying, that is, for π^U restricted to $\{\pi^U < -\pi^O\}$, where the avoided loss is positive realization by realization. The proof in Appendix A.4.5 makes this conditioning explicit.

Private signals about π^U. The unobservable component π^U is drawn from a distribution with mean $\bar{\pi}^U < -\pi^O$ and finite variance σ_π^2. Active protocol users observe noisy private signals about π^U through their usage experience:

$$s_h = \pi^U + \varepsilon_h, \quad \varepsilon_h \sim \mathcal{N}(0, \sigma_h^2), \quad \varepsilon_h \perp \varepsilon_h' \text{ for } h \neq h', \tag{4}$$

where σ_h^2 is the noise in token holder h's private signal. Token holders with low usage intensity (financial investors and passive holders) do not observe informative signals about π^U through usage because they do not actively interact with the protocol; they observe only the common-knowledge component π^O and hold prior beliefs about π^U. The precision of private signals is therefore determined by usage intensity:

Assumption 2 (Usage-based signal precision)**.** *The precision of token holder h's private signal s_h about π^U is strictly increasing in their usage intensity δ_h: $\partial\sigma_h^2/\partial\delta_h < 0$ for $\delta_h > 0$, and token holders with $\delta_h = 0$ do not receive informative signals about π^U.*

Assumption 2 formalizes the natural notion that active protocol users are better positioned to assess the long-run consequences of governance decisions because they directly experience the protocol frictions (higher fees, reduced liquidity, slower response times) that constitute π^U. No amount of financial analytical skill substitutes for direct protocol experience in assessing usage-based value destruction. Importantly, Assumption 2 does not require any individual user to be well-informed about π^U; each signal s_h remains noisy. The aggregate of many such signals is, however, highly informative, which is the role played by discussion forums.

Divergent priors. Beyond signal precision, financially-oriented investors and active protocol users form systematically different prior beliefs about π^U. Financial investors monitor protocols primarily through financial metrics (protocol revenue, total value locked, and token price dynamics), which are informative about π^O but only weakly informative about π^U. Conditional on favorable financial metrics, financial investors form optimistic priors about long-run protocol health. Protocol users, who directly experience protocol quality, form more pessimistic priors about the consequences of extraction.

Assumption 3 (Divergent priors). *Financial investors and active protocol users hold different prior beliefs about the unobservable value component. Formally, $\bar{\pi}^U_W \equiv \mathbb{E}_W[\pi^U] > \bar{\pi}^U_U \equiv \mathbb{E}_U[\pi^U]$, where subscripts W and U denote financial investors and protocol users, respectively.*

Assumptions 2 and 3 together establish that financial investors are doubly disadvantaged in assessing π^U: they hold optimistic priors and receive no informative private signals. The combination generates a systematic bias toward supporting extraction policies that appear financially attractive but are value-destroying once long-run protocol consequences are accounted for. We emphasize that Assumption 3 departs from the common-prior assumption: the disagreement between investors and users is a primitive of the model, not a consequence of differential information. Each type updates by Bayes' rule from its own prior, so both are individually rational given their beliefs, but because the priors themselves differ, the governance failure is not self-correcting through financial analysis alone. We view the divergence in priors as capturing the different information environments the two types inhabit, financial investors monitoring through market metrics and users through direct protocol experience, which we do not model explicitly.

Empirical mapping. The theoretical mechanism operates through the two-dimensional heterogeneity in (θ_h, δ_h), neither of which is directly observable in governance data. Our empirical analysis classifies token holders based on stake size α_h, which serves as an imperfect proxy for the underlying type. We justify this classification with the following assumption, which is weaker than a deterministic mapping:

Assumption 4 (Probabilistic stake-type relationship). *Smaller token holdings are associated, on average, with higher protocol usage intensity and lower financial orientation:*

$$\mathbb{E}[\delta_h \mid \alpha_h \textit{ small}] > \mathbb{E}[\delta_h \mid \alpha_h \textit{ large}] \quad \textit{and} \quad \mathbb{E}[\theta_h \mid \alpha_h \textit{ small}] < \mathbb{E}[\theta_h \mid \alpha_h \textit{ large}]. \tag{5}$$

Assumption 4 allows for substantial overlap between groups (investor-users with large stakes and high usage intensity, or passive holders with small stakes and low usage intensity) and does not require a deterministic relationship between stake size and type. In the empirical analysis below, we provide direct empirical support for Assumption 4: small token holders exhibit significantly higher rates of on-chain protocol interaction than large token holders, and governance outcomes are most strongly predicted by small-holder participation when those holders are confirmed active users. Furthermore, any misclassification of investor-users as financial investors attenuates our estimated effects toward zero, making our empirical results conservative.

4.3 Voting Without Governance Infrastructure

We first characterize voting outcomes in the absence of governance infrastructure, establishing a baseline of financial-investor dominance.

Token holder h votes in favor of the extraction proposal if and only if:

$$\theta_h \cdot \alpha_h \cdot \left(\pi^O + \mathbb{E}_h[\pi^U]\right) > (1 - \theta_h) \cdot \delta_h, \tag{6}$$

where $\mathbb{E}_h[\pi^U]$ is h's posterior belief about the unobservable component given their available information. Participation in governance is individually costly: token holder h incurs a cost $c > 0$ to evaluate the proposal, form a posterior belief, and submit a vote. The participation decision therefore satisfies:

$$B_h > c, \tag{7}$$

where B_h is the expected benefit from participation. For financial investors ($k = 1, \dots, K$), who hold discrete stakes α_k^W, the expected benefit is:

$$B_k^W = \Pr(\text{pivotal}_k) \cdot \left|\theta_k \alpha_k^W \left(\pi^O + \mathbb{E}_k[\pi^U]\right)\right|, \tag{8}$$

where $\Pr(\text{pivotal}_k) > 0$ because financial investors hold non-negligible stakes. For atomistic protocol users ($h \in [0, 1]$), who hold infinitesimal stakes $d\alpha_h^S$, the individual expected benefit is:

$$B_h^S = \Pr(\text{pivotal}_h) \cdot \left|(1 - \theta_h)\delta_h - \theta_h \, d\alpha_h^S \left(\pi^O + \mathbb{E}_h[\pi^U]\right)\right| \approx \Pr(\text{pivotal}_h) \cdot (1 - \theta_h)\delta_h, \tag{9}$$

where $\Pr(\text{pivotal}_h) \to 0$ as the measure of atomistic holders grows large and $d\alpha_h^S \to 0$. Consequently, $B_h^S \to 0$ regardless of the magnitude of δ_h, generating *rational apathy* among atomistic protocol users: even users who correctly assess $\pi^U \ll 0$ and strongly prefer the status quo do not find individual participation worthwhile.

Proposition 1 (Governance capture without infrastructure)**.** *In the absence of governance infrastructure, atomistic protocol users do not participate in equilibrium, and governance outcomes are determined exclusively by financial investors, who vote for the extraction proposal whenever:*

$$\pi^O + \mathbb{E}_k[\pi^U] > 0. \tag{10}$$

Suppose in addition that the financial investor's optimistic prior is insufficiently pessimistic to offset the observable gain,

$$\bar{\pi}_W^U > -\pi^O. \tag{11}$$

Then, since financial investors' posteriors satisfy $\mathbb{E}_k[\pi^U] \approx \bar{\pi}_W^U$ in the absence of a forum signal, condition (10) holds and the extraction proposal passes despite $\Delta V = \pi^O + \pi^U < 0$.

Condition (11) is the substantive content of the governance failure, and it does not follow from Assumptions 1 and 3 alone: Assumption 1 bounds the true mean below $-\pi^O$, while Assumption 3 only orders the two subjective means. Condition (11) additionally requires the investor's prior $\bar{\pi}_W^U$ to lie above $-\pi^O$, which holds when the prior gap $\bar{\pi}_W^U - \bar{\pi}_U^U$ is large enough that investors are optimistic relative to the true mean. This is precisely the case in which financial-investor control produces value destruction.

Proposition 1 establishes the governance failure that governance infrastructure is designed to address: the agents with the most accurate information about π^U, active protocol users, are precisely those with the weakest individual incentives to participate, while financially-oriented investors who systematically underestimate $|\pi^U|$ determine governance outcomes. This is the DAO analog of the free-rider problem in Grossman and Hart (1980): dispersed informed agents cannot individually justify the participation cost, leaving governance to less-informed but financially motivated agents.

4.4 Governance Infrastructure

We now introduce two forms of governance infrastructure, discussion forums and delegation, which mitigate the collective action problem and informational frictions identified above.

Discussion forums. A public discussion forum aggregates individual private signals $\{s_h\}_{h\in\mathcal{A}}$ from participating users into a collective public signal about π^U, where $\mathcal{A} \subseteq [0,1]$ denotes the set of forum participants. When $n = |\mathcal{A}|$ users participate, the aggregate forum signal is:

$$\bar{s} = \frac{1}{n}\sum_{h\in\mathcal{A}} s_h = \pi^U + \frac{1}{n}\sum_{h\in\mathcal{A}} \varepsilon_h. \tag{12}$$

By the law of large numbers, $n^{-1}\sum_{h\in\mathcal{A}} \varepsilon_h \to 0$ as $n \to \infty$, so that $\bar{s} \to \pi^U$. Greater forum breadth therefore improves the precision of the aggregate signal about π^U, with the residual variance $\mathrm{Var}(\bar{s}) = \bar{\sigma}^2/n$ where $\bar{\sigma}^2 = n^{-1}\sum_{h\in\mathcal{A}} \sigma_h^2$ is the average signal noise. For any finite forum the signal is informative but not fully revealing, $\mathrm{Var}(\bar{s}) > 0$, which is what makes the credibility weighting below nondegenerate: agents place weight strictly between zero and one on $\bar{s}$ rather than learning π^U outright.

Discussion forums serve two economically distinct and separately testable roles.

Information aggregation: The forum transforms dispersed private signals, each individually too noisy to be actionable, into a collective public signal that is highly informative about π^U. All token holders observe this signal, but process it differently depending on their ability to verify its content, as we formalize below.

Coordination of delegation: Public forum participation also makes the mass of active, aligned users observable. As we show below, the value of delegating rises with the size of the delegated bloc, so users face a coordination problem: each delegates only when enough others do. By making broad participation against a proposal visible, the forum raises each user's expectation of the bloc size, and hence the value of delegating. The forum therefore acts as a coordination device that facilitates delegation, a channel we formalize through the bloc pivotality $R(n)$ below.

Differential updating. A central feature of our model is that financial investors and protocol users extract different information from the same forum signal $\bar{s}$. A verification friction limits financial investors' ability to learn from the forum.

The friction is a *verification asymmetry.* Active protocol users can validate forum claims against their own private signal s_h: a forum post reporting high usage costs is credible to user h if and only if it is consistent with their own usage experience. Financial investors, who hold no private signal about π^U, have no basis for such verification, and so cannot fully separate genuine usage reports from posts that are noisy or unrepresentative. They therefore place less weight on the forum signal than active users do. This asymmetry is not a matter of financial sophistication: it reflects the irreplaceable informational content of direct protocol experience.

The degree to which financial investors discount the forum depends on the verifiability of its content, which is captured by discussion quality. Higher-quality discussions, those that are more objective, more data-intensive, and more broadly consensual, are easier for an outside observer to corroborate without a private signal, and so narrow the gap between the weight an investor places on the forum and the weight an active user places on it. A related concern, which reinforces the same discounting, is that active users may have incentives to overstate usage costs in order to block proposals they oppose; investors who cannot verify the content rationally allow for this as well. We do not model this incentive formally, as the verification asymmetry alone delivers the credibility wedge we use below.

This friction generates type-specific belief updating rules. For active protocol users:

$$\mathbb{E}_U[\pi^U \mid \bar{s}] = \lambda_U(\text{quality}_i) \cdot \bar{s} + (1 - \lambda_U(\text{quality}_i)) \cdot \bar{\pi}^U_U, \tag{13}$$

and for financial investors:

$$\mathbb{E}_W[\pi^U \mid \bar{s}] = \tilde{\lambda}_W(\text{quality}_i) \cdot \bar{s} + \Big(1 - \tilde{\lambda}_W(\text{quality}_i)\Big) \cdot \bar{\pi}_W^U, \tag{14}$$

where quality_i captures the objective characteristics of forum discussion for proposal i (objectiveness, professionalism, data intensity, and consensus of posts), and $\lambda_U, \tilde{\lambda}_W \in [0,1]$ are the respective credibility weights. We write the quality argument alone for brevity; both weights also rise with forum breadth n, through the precision $\tau_F(n)$ of the aggregate signal $\bar{s}$ in (12), so that $\lambda_U = \lambda_U(\text{quality}_i, n)$ and similarly for $\tilde{\lambda}_W$. Discussion quality governs the gap between them, while breadth raises both. Appendix A.4.3 derives both weights, and Assumption 5, from verifiability fractions $\kappa_U(\text{quality}_i) > \kappa_W(\text{quality}_i)$ that measure the share of the forum's precision each type can corroborate.

Assumption 5 (Differential credibility)**.** *The credibility weights satisfy:*

(i) $\lambda_U(\text{quality}_i) > \tilde{\lambda}_W(\text{quality}_i)$ *for all* quality_i*, reflecting the verification advantage of active protocol users; and*

(ii) $\partial\lambda_U/\partial\text{quality}_i > 0$ *and* $\partial\tilde{\lambda}_W/\partial\text{quality}_i > 0$*, so that higher discussion quality increases credibility for both types, because higher-quality discussions are easier to corroborate without a private signal and more broadly consensual.*

The combination of divergent priors (Assumption 3) and differential credibility (Assumption 5) implies that the same forum signal $\bar{s}$ generates systematically different posteriors across types. When the forum aggregates genuine evidence of value destruction, so that the public signal comes in at or below the financial investor's prior ($\bar{s} \leq \bar{\pi}_W^U$), the two effects reinforce one another: since $\bar{\pi}_W^U > \bar{\pi}_U^U$ and $\tilde{\lambda}_W < \lambda_U$, financial investors remain more optimistic about π^U than protocol users after observing the forum, and this gap persists at all levels of forum quality (we establish this formally in Lemma 2 in the Appendix). The forum therefore improves information more for active protocol users than for financial investors, even though both observe the same public signal. This is the empirically relevant region, since the governance failure we study arises on proposals where deliberation reveals that π^U is strongly negative. However, financial investors do partially update through the $\tilde{\lambda}_W$ channel: the forum is not irrelevant to them, but its influence is attenuated relative to its influence on active users.

Delegation. Delegation allows atomistic token holders to assign their voting power to a designated representative (delegatee) at cost c_{del}, where $c_{\text{del}} \ll c$ is substantially lower than the direct voting cost. Delegation changes the pivotal unit. An individual atomistic holder is never pivotal, but a delegatee who aggregates the voting power of a continuum of delegators holds a non-negligible mass and is pivotal with positive probability. We assume the delegatee internalizes the bloc and votes its members' shared preference, so a user who delegates has their usage-aligned preference carried by a pivotal actor. The relevant object for the delegation decision is therefore the user's private usage value, enjoyed when the bloc prevails and the protocol is protected, weighted by the probability that the bloc is pivotal:

$$B_h^{\text{del}} = \underbrace{\mathbb{E}[\mathbf{1}[\text{bloc is pivotal}]]}_{R(n)} \cdot \Lambda(\text{quality}_i) \cdot (1 - \theta_h)\delta_h > c_{\text{del}}, \tag{15}$$

where $R(n) \equiv \mathbb{E}[\mathbf{1}[\text{bloc is pivotal}]]$ is the probability that the delegatee's pooled voting weight determines the governance outcome. This pivotality rises with the size of the delegated bloc, and, through the coordination role of the forum described above, the bloc is larger when forum participation n is broader; $R(n)$ is therefore increasing in n. The factor $\Lambda(\text{quality}_i)$ is an information-value factor, increasing in discussion quality, capturing that a sharper posterior about π^U raises the perceived value of carrying one's usage preference into the outcome. As with the credibility weights, we write the quality argument alone for brevity; Λ also rises with breadth through $\tau_F(n)$, and Appendix A.4.4 sets it equal to the user's post-forum posterior precision. Because the pivotality is governed by the bloc's aggregate mass rather than the individual's infinitesimal stake, B_h^{del} is bounded away from zero, unlike the direct-voting benefit in (9). Delegation does not make the individual decisive; it makes the entity that represents them decisive, and the low cost c_{del} then makes participation worthwhile for users with sufficiently high usage stakes. The benefit from delegating is thus increasing in the mass of other users who also delegate, a coordination complementarity that the forum resolves by making broad participation observable.

Delegation and discussion forums are complementary governance mechanisms. Forums aggregate information and make the stakes of a proposal salient, increasing the perceived benefit of participation and therefore the incentive to delegate. Delegation translates dispersed user preferences into effective governance power, amplifying the impact of informed participation facilitated by the forum. We formalize this complementarity in Proposition 2.

4.5 Equilibrium Predictions

We now derive the model's four main predictions. Formal proofs are provided in the Appendix; here we state results and provide economic intuition.

Proposition 2 (Governance infrastructure and participation)**.** *The equilibrium participation rate of protocol users is strictly increasing in:*

(i) *forum participation n, through the coordination channel: broader participation coordinates a larger delegated bloc, raising its pivotality $R(n)$;*

(ii) *forum quality* quality_i*, through the information aggregation channel in (13): higher quality increases λ_U, sharpening users' posteriors about π^U and raising the perceived benefit of participation; and*

(iii) *the enabling of delegation, which reduces the effective participation cost from c to $c_{\text{del}} \ll c$.*

These effects are complementary: the participation gain from delegation is strictly increasing in forum quality quality_i*, and the participation gain from forum breadth is strictly larger when delegation is enabled.*

Proposition 2 yields distinct empirical predictions for the coordination and information aggregation channels. The coordination channel implies that forum breadth, captured empirically by the number of discussants and the concentration of posts (low HHI of posts), should predict small-holder participation. The information aggregation channel implies that forum quality, captured by LLM-based measures of objectiveness, data intensity, and professionalism, should predict participation beyond what breadth alone explains. The delegation effect predicts a discrete increase in participation upon enabling delegation. The complementarity implies that the interaction of delegation with forum quality should be positive in explaining participation rates.

Proposition 3 (Governance infrastructure and user influence)**.** *The probability that governance outcomes reflect the preferences of informed protocol users is strictly increasing in forum participation n, forum quality* quality_i*, and the enabling of delegation. Formally, there exists a threshold usage intensity $\bar{\delta}(n, \text{quality}_i, c_{\text{del}})$ such that users with $\delta_h > \bar{\delta}$ participate under governance infrastructure but not without it, where $\bar{\delta}$ is strictly decreasing in n and* quality_i *and strictly increasing in c_{del}. As $\bar{\delta}$ falls, the composition of the participating voter pool shifts toward users with higher usage intensity, those with the most precise private signals about π^U, increasing the probability that governance outcomes reflect the true value-destroying nature of the extraction proposal.*

Proposition 3 establishes that governance infrastructure shifts not only the quantity but also the quality of participation: the marginal participant enabled by governance infrastructure has higher usage intensity δ_h than the average non-participant, shifting the composition of the voter pool toward better-informed agents. The threshold $\bar{\delta}$ formalizes the marginal user whose participation becomes individually rational under governance infrastructure, and its dependence on all three infrastructure dimensions captures the complementarity between forums and delegation established in Proposition 2.

Proposition 4 (Valuation effects)**.** *Conditional on extraction proposals being value-destroying (Assumption 1), governance outcomes in which informed protocol users successfully block extraction are associated with positive cumulative abnormal returns:*

$$\text{CAR} = \mathbb{E}\big[|\pi^U| - \pi^O \mid \mathcal{I}_{\text{win}}\big] > 0, \tag{16}$$

where $\mathcal{I}_{\text{win}}$ denotes the event of a user-preferred governance outcome. The magnitude of the abnormal return is strictly increasing in forum quality:

$$\frac{\partial\, \text{CAR}}{\partial\, \text{quality}_i} > 0. \tag{17}$$

The intuition for Proposition 4 is as follows. Long-horizon market participants correctly understand that $\Delta V < 0$ when $|\pi^U| > \pi^O$ but cannot directly observe π^U. When informed users block extraction, the status quo is preserved and the market prices the avoided loss. The positive sign in (16) follows directly from Assumption 1: on the relevant class of proposals the avoided loss $|\pi^U| - \pi^O$ is positive for every realization, so its expectation is positive conditional on a user victory. The result does not rely on a user victory selecting more value-destroying proposals; it holds for the avoided loss itself.

The amplification with forum quality in (17) operates through the long-horizon participants' assessment of how large that avoided loss is. Being rational but unable to observe π^U directly, they price the avoided loss using a posterior formed from the public forum signal. As for the agents in the model, they apply a market credibility weight $\lambda_M(\text{quality}_i)$ to the forum signal, increasing in quality by the same verification logic, so they place more weight on it when discussion quality is higher. When the forum reveals value destruction beyond what the prior already reflects, greater credibility pulls their assessment toward the large signal and away from the prior, raising the priced avoided loss and hence the abnormal return. Higher forum quality therefore amplifies the market reaction to a user-preferred outcome.

4.6 Discussion

Our framework contributes to the governance literature along three dimensions.

First, we establish a complementary information structure in which usage experience, not financial sophistication, is the relevant private information for assessing long-run governance consequences. This inverts the standard assumption of the corporate governance literature, in which large shareholders are the informed party (Shleifer and Vishny, 1986; Maug, 1998; Grossman and Hart, 1980). The inversion is natural in the DAO context, since protocol users possess irreplaceable firsthand knowledge about how governance decisions affect protocol quality, but has broader implications for any governance setting in which the consequences of decisions are experienced differentially across stakeholders with varying degrees of direct exposure.

Second, we provide distinct theoretical foundations for two specific governance mechanisms, and identify the channel through which each operates. Discussion forums improve governance through both an information aggregation channel, which makes the unobservable component π^U more transparent, and a coordination channel, which enables users with aligned preferences to act collectively. These two channels generate separately testable predictions: information aggregation is identified by forum quality measures (objectiveness, data intensity), while coordination is identified by forum breadth measures (participation counts, post concentration). Delegation improves governance by lowering participation costs and enabling collective action, and its effectiveness is complementary to forum quality. These distinctions refine the existing literature on shareholder coordination (Levit and Malenko, 2011; Malenko and Malenko, 2019), which typically treats participation as undifferentiated, by showing that the quality of participation, not merely its quantity, determines governance outcomes.

Third, our market valuation result in Proposition 4 connects governance process quality to asset prices through a specific and testable channel. The positive CAR associated with user-preferred governance outcomes reflects the market's inference about the magnitude of π^U avoided, and its amplification with forum quality reflects the greater informativeness of victories achieved through credible information aggregation. This prediction is directional and conditional on Assumption 1: it applies to value-destroying proposals, not to all contested governance events.

These features jointly generate four empirical hypotheses:

(H1) Governance infrastructure increases participation among active protocol users (Proposition 2);

(H2) Governance infrastructure increases the probability that informed users determine governance outcomes (Proposition 3);

(H3) User-preferred governance outcomes on value-destroying proposals are associated with positive abnormal token returns (Proposition 4); and

(H4) The magnitude of valuation effects is increasing in the quality, not merely the presence, of governance infrastructure (Proposition 4).

5 Sample and Data

This section describes the data sources used in our analyses and the procedures we use to construct the proposal-, voter-, and discussion-level variables. Table A.1 in the appendix provides an overview of all variables.

5.1 Sample

Our sample consists of governance proposals from DAOs that govern DeFi protocols on Ethereum. We obtain the list of DAOs from Snapshot, the leading off-chain voting infrastructure for DAOs, and match each space to its governance token on CoinGecko. We restrict the sample to proposals whose voting end date falls on or before September 1, 2025, and whose governance token has price data spanning the event-study estimation window. Proposals associated with mixed-type or staking-derivative governance tokens that cannot be cleanly mapped to a single underlying token are excluded. The final sample contains 2,830 proposals.

5.2 Proposal Data

We obtain proposal metadata and ballot-level vote data from the Snapshot GraphQL endpoint. For each proposal i, we collect the proposing address, the creation, voting start, and voting end timestamps, the set of choices, the voting mechanism (single choice, weighted, ranked choice, and multiple choice), the quorum requirement, and the delegation rule. At the vote level, we record the voter address, the chosen alternative, the voting power, the timestamp, and the optional reason text. We construct $\mathit{Weighted}_i$, $\mathit{MultipleChoices}_i$, Quorum_i, and $\mathit{Delegation}_i$ as dummies for the corresponding mechanism features, and $\mathit{Discussion}_i$ as a dummy that equals one if the proposal has a non-empty discussion-link field on Snapshot.

We also obtain each DAO's self-declared Snapshot space category (e.g., *defi*, *investment*, *social*). When a space declares multiple categories we use the first listed, and we assign *other* when no category is declared. Besides, we create indicators that classify proposals into the following not-mutually-exclusive topics: protocol security, treasury expenditure, user incentive, tokenomics, and voting mechanism.

5.3 Token Holding and Transaction Data

We obtain governance token balances from Ethereum archive nodes accessed through Infura. For each proposal i, we recover the set of holders at the snapshot block underlying the proposal and define $\#\ \mathit{Holders}_i$ as the number of EOAs with a non-zero balance. For DAOs whose governance token is held in a separate staking or vote-escrow contract, we aggregate balances across the corresponding contracts before classifying holders. We further compute the share of circulating supply held by each address and classify a holder as a *whale* if she holds at least 5% of supply and as a *small token holder* otherwise.

For each voter h, we link the voting address to its full Ethereum transaction history, obtained from a Flipside Snowflake archive node. We classify a voter as a user if her address has previously interacted with the smart contract of the DeFi protocol governed by the proposing DAO. For each proposal, we calculate the total number of users ($User_i$) and separately compute this variable ($User_i^{Small}$) for the subset of small-token-holder voters.

We also record each voter's governance token transfers to and from known CEX and DEX addresses in the $[-5, +5]$ day window around the voting end date. CEX and DEX addresses are identified using the labelled-address table maintained by Flipside. For each (proposal, voter, day) triple, $Buy_{i,h,t}$ counts inbound transfers from CEX/DEX addresses and $Sell_{i,h,t}$ counts outbound transfers to CEX/DEX addresses.

5.4 Discussion Data

DAO governance proposals are typically preceded by an off-chain discussion thread on a Discourse forum. We scrape the full discussion history for each DAO from its public forum URL and match threads to Snapshot proposals using the discussion-link field. For forums that block automated requests, we download the source HTML.

For proposals with at least two discussion posts, we use GPT-4o to evaluate whether the discussion thread exhibits consensus. The model is asked to return a Boolean judgment for each thread. To obtain a continuous measure we retrieve the log-probability of the Boolean output token and convert it to the implied probability that the thread exhibits consensus, which is bounded in $[0, 1]$.

For each proposal we evaluate consensus on two configurations of the reply sequence: (*before*), the first half of the replies in chronological order, and (*full*), the entire reply sequence. We define

$$\Delta Consensus_i = Consensus_i^{full} - Consensus_i^{before}, \tag{18}$$

which captures the change in the discussion consensus measure as the discussion is extended from the first half to the entire thread. We use the full thread rather than the second-half segment alone in the comparison so that the model evaluates consensus over the complete discussion, preserving the context of the early replies that later participants themselves had when contributing.

5.5 Token Returns

We obtain daily governance token prices from CoinGecko and daily market returns from a value-weighted cryptocurrency index that we construct from CoinGecko market caps. The risk-free rate is the daily three-month U.S. Treasury bill rate.

For each proposal i, we compute ${CAR^{Create}}_i$ and ${CAR^{End}}_i$ as the cumulative abnormal return of the governance token over the $[-5, +5]$ trading-day window centered on the proposal creation and voting end dates, respectively. Expected returns are estimated using the CAPM with an estimation window of 250 to 20 trading days prior to the event. We require each event to have non-missing price data over both the estimation window and the event window.

5.6 Voting Outcomes

For each proposal i, we compute the majority choice separately for small token holders and for whales, weighting each voter by vote count. We classify proposal i as a small-token-holder victory and set $SmallWin_i = 1$ if the final voting outcome matches the majority choice of small token holders but differs from the majority choice of whales. For each (proposal, voter) pair, $Against\ Outcome_{i,h,t}$ equals one in days following the voting end date if voter h cast a vote that differs from the final outcome of proposal i, and zero otherwise.

5.7 Descriptive Statistics

Table 2 reports summary statistics for the key variables that enter our analyses. Continuous variables are winsorized at the 1% and 99% levels. Specifically, the mean cumulative abnormal return is -1.8% over the $[-5, +5]$ window around proposal creation and -4.0% around voting end. The average participation for small token holders is low at 3.2%. Yet, among 2,669 proposals containing small token holders, 181 (6.8%) are classified as small-token-holder victories. Among the mechanism features, 13.4% of proposals adopt the weighted voting mechanism, 42.3% allow multiple choices, 29.9% impose a quorum, 26.4% enable delegation, and 39.5% have an associated Discourse discussion thread. Conditional on the proposals with discussion threads, the change in consensus is on average positive at 0.136.

6 Research Design

This section describes the empirical specifications used in our following analyses. We examine (i) the determinants of small token holder participation, delegation, and victories, (ii) the causal effect of enabling delegation on participation, (iii) the market reaction to small token holder victories, and (iv) post-vote trading behavior. Because governance on a public blockchain is observable at the account level—we see the complete ballot for each proposal, the off-chain deliberation that precedes it, and each voter's token transfers around the event, and we can link a voting address to its full transaction history—the specifications below identify behavior at both the proposal level and the individual voter level.

6.1 Determinants of Small Token Holder Participation, Delegation, and Victories

We examine how proposal characteristics, off-chain discussion, and protocol-user engagement relate to small token holder governance outcomes. For each proposal i at calendar year t, we estimate

$$Y_{i,t} = \alpha + \boldsymbol{\beta}'\mathbf{X}_{i,t} + \mu_c + \tau_t + \varepsilon_{i,t}, \tag{19}$$

where $Y_{i,t}$ is one of $\{Participation_{i,t}^{Small}, Delegation_{i,t}^{Small}, SmallWin_{i,t}\}$, $\mathbf{X}_{i,t}$ is a vector of proposal characteristics that includes the voting mechanism dummies, the discussion measure, and the protocol-user engagement indicator, μ_c are DAO-category fixed effects, and τ_t are year fixed effects.

We estimate (19) for $Y_{i,t} = Participation_{i,t}^{Small}$ and $Y_{i,t} = Delegation_{i,t}^{Small}$ by ordinary least squares with high-dimensional fixed effects. Following the methodology outlined by Anderson et al. (2018) and Agrawal et al. (2022), we employ a panel logistic Poisson regression with fixed effects to estimate the small token holder victory equation, which handles the binary outcome and the DAO-category fixed effects in a unified framework.

For each outcome we report three specifications. Column (1) reports the baseline on the full sample. Column (2) restricts the sample to proposals with at least two discussion posts and replaces $Discussion_{i,t}$ with $\Delta Consensus_{i,t}$. Column (3) restricts the sample to DAOs whose protocols have a deployed Dapp and augments the baseline with $User_{i,t}^{Small}$. We report heteroskedasticity-robust standard errors for the participation and delegation specifications and cluster the standard errors at the DAO-category level for the small token holder victory specification.

6.2 Difference-in-Differences: Enabling Delegation

A substantial fraction of DAOs adopt delegation at different points in time over our sample. To estimate the causal effect of enabling delegation on participation, we exploit this staggered adoption in a stacked difference-in-differences design that follows Cengiz et al. (2019) and Deshpande and Li (2019). The stacked design mitigates the bias from heterogeneous treatment effects under staggered adoption documented by Goodman-Bacon (2021), Callaway and Sant'Anna (2021), and Sun and Abraham (2021).

For each treated DAO that ever enables delegation, let t_i denote the year of adoption. We group all treated DAOs with $t_i = t_c$ into a cohort c, and match each cohort to a control panel of DAOs that never enable delegation over the sample. We stack the cohorts and estimate

$$Participation^{s}_{c,i,t} = \beta \cdot Treat_{c,i} \times Post_{c,t} + \delta_{c,i} + \delta_{c,t} + \varepsilon_{c,i,t}, \tag{20}$$

where $s \in \{Small,\ Whale\}$, $Treat_{c,i}$ equals one if the DAO ever enables delegation within cohort c and zero otherwise, $Post_{c,t}$ equals one for proposals that occur after delegation is enabled, and $\delta_{c,i}$ and $\delta_{c,t}$ are cohort-DAO and cohort-year fixed effects. The coefficient β captures the average treatment effect on the treated of enabling delegation on participation. The standard errors are clustered at the DAO level.

We complement the static specification with an event-study analogue that replaces $Treat_{c,i} \times Post_{c,t}$ with a vector of leads and lags relative to the adoption date. This event-study specification allows us to test the parallel-trends assumption that underlies (20) and to trace the dynamic profile of the treatment effect.

6.3 Market Reaction

For each proposal i, we compute daily abnormal returns over the $[-5, +5]$ trading-day window centered on either the proposal creation date or the voting end date. Expected returns are estimated using the CAPM on an estimation window of 250 to 20 trading days prior to the event. We then accumulate the abnormal returns over the event window to obtain ${CAR^{Create}}_i$ and ${CAR^{End}}_i$. To examine the market reaction to proposals in which small token holders prevail, we estimate

$$CAR^{s}_{i,t} = \alpha + \beta \cdot SmallWin_{i,t} + \mu_k + \tau_t + \varepsilon_{i,t}, \tag{21}$$

where $s \in \{Create,\ End\}$, μ_k are token fixed effects, and τ_t are year fixed effects. The standard errors are clustered at the token level.

To test whether this valuation effect varies with the quality of the deliberation that preceded the vote, we restrict the sample to proposals with at least two discussion posts, split it at the average of $\Delta Consensus_i$, and estimate (21) separately in the below- and above-average subsamples. A sample split avoids imposing a functional form on the quality gradient and is the cleaner test given the small number of small-holder victories for which discussion quality is defined.

6.4 Post-Vote Trading

Finally, we examine whether small token holders trade the governance token following the conclusion of a proposal. The account-level transfer data are central to this test: because we observe transfers at the voter-day level, we can relate a voter's post-vote trading to that voter's own position in the proposal—whether the voter was on the winning side—a margin that an aggregate price reaction cannot identify. For each voter h in proposal i, we observe the number of inbound transfers from CEX/DEX addresses ($Buy_{i,h,t}$) and the number of outbound transfers to CEX/DEX addresses ($Sell_{i,h,t}$) on each day t in the $[-5, +5]$ window around the voting end date. We estimate

$$Trade^{s}_{i,h,t} = \beta \cdot Z_{i,h,t} + \delta_k + \tau_t + \varepsilon_{i,h,t}, \tag{22}$$

where $s \in \{Buy,\ Sell\}$ and $Z_{i,h,t}$ is either $SmallWin_{i,t}$ or $Against\ Outcome_{i,h,t}$. Following the methodology outlined by Anderson et al. (2018) and Agrawal et al. (2022), we employ a panel logistic Poisson regression with fixed effects to estimate (22). When the regressor of interest is $SmallWin_{i,t}$, we absorb voter and year fixed effects and cluster the standard errors at the voter level. When the regressor of interest is $Against\ Outcome_{i,h,t}$, we absorb DAO-category and year fixed effects and cluster the standard errors at the DAO-category level.

7 Results

In this section we present our empirical findings. We proceed in five steps. We begin with univariate evidence, splitting the sample first by holder size and then by whether delegation is enabled to establish the basic facts. We then use regression approaches to examine what determines small token holder participation, delegation, and influence over outcomes. Next we isolate the causal effect of enabling delegation on participation using its staggered adoption. We then trace how small token holders win when they do, and finally we examine the market reaction to their victories and the trading that follows.

7.1 Univariate Analysis

We begin with a univariate split of the sample that establish the facts the model is built to explain: that small token holders are dispersed, apathetic, but better informed than whales.

Table 3 compares small token holders (below 5% of supply) and whales (5% or more) along the holding and participation margins aggregated to the proposal level. Three facts stand out. First, the two groups differ enormously in number: an average proposal has roughly 6,500 small holders against fewer than three whales, so the small-holder side is the dispersed population and the whale side a handful of large blocks. Second, small holders are far less engaged along every participation margin. Their turnout is 2.6% against 52.9% for whales, their delegator rate is 3.5% against 32.7%, and they supply reasons for their votes far less often (0.9% against 4.5%). This is the empirical counterpart of the rational apathy result in Proposition 1: an atomistic holder is close to non-pivotal, so individual participation is not worthwhile, whereas a whale holds a decisive block and participates. Third, and cutting the other way, small holders are *more* likely than whales to have used the protocol they govern: the mean of $User_i$ is 0.68 for small holders against 0.56 for whales. The dispersed, apathetic side is therefore also the better-informed side—the inversion at the center of the model.

7.2 Determinants of Small Token Holder Participation, Delegation, and Victories

Participation. Proposition 2 predicts that the participation of protocol users is increasing in the breadth of forum discussion, the quality of that discussion, and the availability of delegation. Table 4 reports estimates of equation (19) for $Participation^{Small}_{i,t}$.

The estimates support all three channels and map onto the two distinct roles that forum discussion plays in the model. Enabling delegation raises small token holder participation by 2.3 percentage points (column 1), a large effect relative to the sample mean of 3.2%, and the presence of a discussion thread raises it by 0.8 percentage points. These two coefficients identify the *coordination* channel of Proposition 2: forum breadth makes the stakes of a proposal salient and the intentions of like-minded holders observable, and delegation lowers the cost of acting on that information. In the model, breadth operates through the law-of-large-numbers aggregation of dispersed signals and through the pooled pivotality of the delegated bloc—channels governed by the number of participants rather than by the content of what they say. The *information aggregation* channel is identified separately in column 2, which restricts the sample to proposals with at least two discussion posts and replaces the discussion dummy with $\Delta Consensus_{i,t}$, the change in the consensus of the thread as it unfolds. The coefficient on $\Delta Consensus_{i,t}$ is positive and significant, so participation responds not only to the presence of discussion but to its informational content. This is the distinction the model draws between forum breadth and forum quality: in the framework these enter through *separate* primitives—breadth through the aggregate signal precision $\tau_F(n)$ and the bloc pivotality $R(n)$, quality through the credibility with which that signal is weighted—so that our count-based and consensus-based measures identify non-overlapping mechanisms rather than proxying for one another.

Column 3 restricts the sample to DAOs whose protocols have a deployed Dapp and adds

$User^{Small}_{i,t}$, the share of small token holder voters who have previously interacted with the protocol. The coefficient is positive and significant: participation is highest where small token holders are confirmed protocol users. This is direct evidence for the mechanism of the model—usage intensity, not stake size, is the source of the private information that governance infrastructure mobilizes (Assumption 2), and it is the holders with that information who respond most. The coefficient on $Weighted_{i,t}$ is negative throughout, confirming that voting mechanisms which concentrate voting power dampen small-holder participation.

Delegation. Table 5 repeats the exercise for $Delegation^{Small}_{i,t}$, the share of small token holders who delegate. The pattern mirrors the participation results: the presence of discussion, the change in consensus, and confirmed protocol usage all raise delegation, while the voting-mechanism dummies are insignificant. Delegation and discussion therefore operate as complementary forms of governance infrastructure, as Proposition 2 predicts: discussion raises the perceived value of participation, and delegation provides a low-cost channel through which dispersed users act on it. This is the empirical reflection of the supermodularity of the participation benefit established in the proof of Proposition 2—the marginal effect of broader discussion on participation is larger when delegation is available.

Which proposals draw small holders in. The model implies that protocol users hold private information about the long-run consequences of governance for protocol quality, so they should participate more in proposals that bear directly on the protocol they use. Table 6 reports participation by proposal topic. The pattern is consistent with this reading: small token holder participation is higher for proposals concerning protocol security and for proposals that increase user incentives, the two topics most directly tied to the user experience, and does not respond significantly to treasury, tokenomics, or voting-mechanism proposals. Whale participation is unrelated to topic throughout. This selective engagement is the empirical signature of the usage-based information channel: small holders participate where their private signal about protocol quality is relevant. Consistent with this, we do not find that ballot complexity deters small holders—the coefficient on $MultipleChoices_{i,t}$ in Table 4 is small and insignificant. What discourages their participation is not the number of options on the ballot but the concentration of voting power, since $Weighted_{i,t}$ enters negatively and significantly.

Victories. Proposition 3 predicts that governance infrastructure increases the probability that outcomes reflect the preferences of informed users, and that the marginal participant it draws in has higher usage intensity and hence a more precise signal. Table 7 reports panel logistic

Poisson estimates of how proposal characteristics relate to the likelihood of a small token holder victory, defined as an outcome that matches the majority choice of small token holders but differs from that of whales.

The results align with the participation evidence. Enabling delegation raises the likelihood of a small token holder victory (column 1), and the effect is robust to restricting the sample to DAOs with a deployed Dapp (column 3). The change in discussion consensus is positive and significant in column 2, so higher-quality deliberation predicts not only participation but also influence over the final outcome. Most directly, the share of confirmed protocol users among small token holders, $User_{i,t}^{Small}$, is a strong positive predictor of victories. This is the empirical content of Proposition 3: as governance infrastructure lowers the participation threshold $\bar{\delta}$, the marginal participant is a higher-intensity user with a more precise signal (Assumption 2), and the composition of the voter pool shifts toward the better-informed agents who carry the outcome.

The coefficient on $Weighted_{i,t}$ is positive and large, which may appear to contradict its negative effect on participation in Table 4, but the two are consistent. Weighted voting suppresses the number of small holders who participate, yet conditional on a contested proposal reaching the ballot under weighted voting, the cases in which small holders prevail are concentrated among proposals where they coordinate intensely. We read this coefficient with caution and do not interpret it as evidence that concentrating voting power benefits small holders.

7.3 The Causal Effect of Enabling Delegation

The cross-sectional estimates above are informative but cannot rule out that DAOs which enable delegation differ systematically from those that do not. To isolate the effect of enabling delegation on participation, we exploit its staggered adoption across DAOs in the stacked difference-in-differences design of equation (20).Table 8 reports the results.

Enabling delegation raises small token holder participation by 1.5 percentage points (column 1), a statistically significant effect that is economically meaningful against the baseline mean of 3.2%. The effect on whale participation (column 2) is positive but insignificant. Delegation thus closes part of the participation gap between small holders and whales, in line with Proposition 2: by lowering the participation cost from c to $c_{\text{del}} \ll c$, it has its largest effect on the holders for whom the cost was previously binding—the dispersed small holders, not the whales who were already participating. Figure 2 plots the dynamic treatment effects. The pre-adoption coefficients are close to zero and statistically insignificant, supporting the parallel-trends assumption, and the effect emerges only after delegation is enabled.

7.4 How Small Token Holders Win

Small token holder victories are infrequent—of the 2,830 proposals, 181 (6.4%) are victories by vote count, 61 (2.2%) by voting power, and 56 (2.0%) by both. On most proposals concentrated holders determine the result, as the prior literature documents. It is precisely because these victories are rare, yet valuable when they occur, that we study how they come about. A rare outcome that nonetheless arises systematically under particular conditions reveals what those conditions are. We use the account-level data to trace the channel through which dispersed votes become a decisive bloc, asking first *through what* small holders participate when they win, and then *with whose* votes.

Through delegation, not direct voting. The victories are concentrated where governance infrastructure is present: 170 of the 181 occur in DAOs that enable delegation, where the victory rate is 24.9%, against 0.55% in DAOs without delegation. Delegation is therefore close to a necessary condition for small holders to prevail, consistent with the participation mechanism in Proposition 2: without a low-cost channel to pool their votes, dispersed holders do not overcome the collective action problem. This is the empirical counterpart of the pivotality argument in the model—an individual atomistic holder has pivotality near zero, whereas the delegated bloc has pivotality $R(n)$ bounded away from zero, so participation becomes worthwhile only once votes can be pooled. Table 9 decomposes small holder participation into direct-voting and delegated components for delegation-enabled proposals. Across all such proposals, delegation accounts for 58% of small holder participation. Conditional on a victory the reliance on delegation is higher still: the direct-voting share falls to 1.5% while the delegated share rises to 5.1%, so that 77% of the participation behind a victory is delegated. Small holders win mainly by delegating their votes to representatives, not by voting individually.

By pooling their own power, not allying with whales. A natural question is whether small holders prevail on their own or only with the support of some larger holders. We reconstruct the voting power cast on each side of every victory proposal and classify each voter by supply share into small token holders (below 5%), small whales (5 to 10%), and large whales (10% or more). Table 10 reports the average composition of the winning and losing sides. Small holders win on their own power: on the winning side of a vote-count victory, 96.9% of the voting power comes from small holders, and in 95.5% of these victories no whale appears on the winning side at all. Whales are predominantly on the losing side—large whales cast 79% of their voting power against the small-holder outcome, and small whales do the same. The pattern is robust to defining victories by voting power rather than vote count, where small holders still supply 94.8% of the winning side. Small holder victories are therefore not coalitions with sympathetic whales against larger ones; they are instances in which dispersed small holders pool enough of their own voting power, through the delegation channel documented above, to outweigh the whales who oppose them. This is the empirical content of the collective action mechanism in Propositions 2 and 3: the governance infrastructure that mobilizes small holders lets their aggregated voice, rather than a realignment among large holders, determine the outcome.

7.5 The Value of Small Token Holder Victories

Having established that small holders win by pooling their own votes through delegation, we now ask what those victories are worth. We examine the market reaction to a victory and then the trading behavior of the holders who produced it.

Market reaction. Proposition 4 predicts that, conditional on extraction proposals being value-destroying, governance outcomes in which informed users block extraction are associated with positive cumulative abnormal returns. Before conditioning on victories, we show that the average proposal earns a negative reaction, consistent with the prior literature. Figure A.1 in the Appendix reports cumulative abnormal returns (CARs) over the event window and is slightly negative on average, so governance events are value-relevant but the typical proposal is met with a slightly negative market response. In contrast to this baseline, Table 11 reports estimates of equation (21). A small token holder victory is associated with a 4.6% higher CARs around proposal creation and 4.5% around voting end, both statistically significant and economically large. The market therefore treats a small holder victory as good news, consistent with Proposition 4: when dispersed users overcome the collective action problem and block a proposal, the outcome reveals that the avoided value destruction was large, and long-horizon investors price this in. Figure 3 traces the dynamics; the abnormal return accumulates around both event dates, and the gap from other proposals is concentrated in the days surrounding the event rather than present beforehand.

One caveat applies to the mapping between the model and this test. The model's valuation prediction is stated for the value-destroying class of proposals—those satisfying $|\pi^U| > \pi^O$ in Assumption 1, on which Proposition 4 is established—whereas $SmallWin_{i,t}$ identifies any contested proposal whose outcome matches the small-holder majority but not the whale majority. Not every small holder victory is the blocking of a value-destroying proposal, so $SmallWin_{i,t}$ is a noisy proxy for the model's object. To the extent that some victories are not extraction-blocking events, this works against finding a positive abnormal return, so our estimate is conservative.

The model also predicts that the valuation effect is increasing in the quality of deliberation, not merely its presence (Proposition 4, eq. (17)). Table 13 tests this by splitting proposals with at least two discussion posts at the average level of $\Delta Consensus_{i,t}$ and re-estimating the Table 11 specification in the low- and high-quality subsamples.

Around voting end, the coefficient on $SmallWin_{i,t}$ rises from 0.055 (significant at 1%) in the low-quality subsample to 0.081 (significant at 10%) in the high-quality subsample: a small-holder victory preceded by higher-consensus deliberation earns a larger abnormal return, the direction Proposition 4 predicts. Around proposal creation the coefficient is essentially flat (0.042 to 0.044) and only crosses conventional significance in the high-quality subsample. This contrast across windows is itself informative: the creation-window return is taken before the vote's outcome is known and so reflects whatever the forum signal has already been impounded into prices, whereas the end-window return resolves the residual uncertainty about whether extraction was blocked and is the cleaner test of the credibility channel in the model. The quality gradient is sharpest exactly where the model says it should be, and essentially absent where the model says it need not appear, which is consistent with Proposition 4 rather than incidental to it.

Post-vote trading. We now turn to how small holders trade after a proposal concludes. Figure 4 first summarizes the unconditional flows of whales and small token holders to and from centralized and decentralized exchanges around the proposal creation and voting end dates. Both groups trade actively, and small token holders sell somewhat more than whales around the voting-end window. This aggregate comparison, however, cannot speak to whether a holder's trading depends on the outcome aligned with her preferences. The account-level structure of our data lets us go further: the unit of observation is the voter-day, so we observe the buy and sell transfers of each individual voter around the vote and can condition on whether that voter was on the winning side. An aggregate event study, of the kind used to study corporate shareholder votes, cannot make this distinction; it observes the price reaction but not the trades of the participants who produced it. If governance token positions were held for speculative reasons, holders on the winning side would be expected to sell into the positive reaction once the outcome is known; if instead small holders hold tokens because they value the protocol they use, a victory should not trigger selling. Table 12 reports panel logistic Poisson estimates of equation (22) for buy and sell volume around the voting end date.

Small token holders sell *less* following a victory: the coefficient on the victory indicator in the sell regression is negative and significant, while the corresponding buy coefficient is insignificant. Disagreement with the final outcome, captured by $Against\ Outcome_{i,h,t}$, is unrelated to either buying or selling. Small holders who prevail therefore retain their tokens rather than sell into the positive market reaction documented in Table 11. This is consistent with the model's premise that the participating small holders are protocol users whose holdings reflect long-run alignment with protocol health—a low financial orientation θ_h and a positive usage stake δ_h—rather than financial investors trading on short-run price movements.

8 Conclusion

In this paper, we study whether small shareholders have voice in DAO governance and whether that voice creates value. We develop a model in which token holders differ in the weight they place on financial returns relative to protocol utility and in the intensity with which they use the protocol. Active users observe private signals about the long-run consequences of a governance decision through their usage, while financial investors observe only short-run financial metrics and hold more optimistic priors. Without governance infrastructure, the informed users are non-pivotal and do not participate, so financial investors determine outcomes and value-destroying proposals pass.

The model shows how two features of DAO governance relax this friction. Discussion forums aggregate the dispersed private signals of users into a public signal and coordinate users with aligned preferences, and delegation lowers the cost of participation and lets dispersed users pool their voting power. These mechanisms raise the participation of informed users, shift the composition of participants toward those with more precise information, and raise the probability that governance outcomes reflect the preferences of users rather than financial investors.

The evidence is consistent with these predictions. Small token holders are far less engaged than whales but are more likely to use the protocols they govern. The availability of delegation, the presence and consensus of discussion, and confirmed protocol usage all raise small token holder participation, while weighted voting lowers it, and the staggered adoption of delegation raises small token holder participation in a difference-in-differences design. The same features raise the probability of a small holder victory. When small token holders win, they win by pooling their own votes through delegation rather than by allying with sympathetic large holders: they supply about 97% of the winning side, most of it through delegated votes, while whales are predominantly on the losing side. Small holder victories are associated with positive abnormal returns, against a mildly negative reaction for proposals on average, and this valuation effect is larger when the deliberation preceding the vote reached more consensus. Small token holders who prevail hold their tokens rather than sell into the positive reaction, consistent with their holdings reflecting long-run alignment with the protocol.

Our results speak to the governance of dispersed organizations broadly, not only to DAOs. The corporate governance literature has largely treated small token holders as uninformed and rationally apathetic, and large blockholders as the informed monitors. We find that the opposite can hold. When small token holders are the parties who directly experience the consequences of a decision, their dispersed and individually noisy signals carry information that large holders do not have. Their voice is then worth hearing, and a governance scheme that aggregates it can improve decisions rather than ceding control to the largest and most financially motivated holders. The deliberation forum and delegation we study are one way to aggregate that voice; the broader lesson is that the value of small-holder voice depends on the institutions available to aggregate it.

The forces we identify are visible outside decentralized finance. The GameStop episode of 2021, in which dispersed retail investors coordinated through an online forum against concentrated institutional short positions, is a prominent instance of small, dispersed participants acting collectively against concentrated financial interests. Our setting differs in an instructive way. The contest we study takes place through voting on governance proposals rather than through trading, and the small holders who prevail are the informed users of the protocol who hold their tokens rather than sell, so their coordination reflects long-run alignment rather than short-run speculation. Giving dispersed but informed stakeholders the tools to coordinate can improve governance, and need not amount to a speculative frenzy.

The same logic applies to traditional organizations. In a large corporation, the informed but dispersed parties may be the customers, employees, or small token holders who interact with the firm's products and operations, and who individually lack the incentive to become active. Our findings suggest that mechanisms which lower the cost of participation and aggregate dispersed information, such as structured deliberation and delegation, can give this voice weight and make governance more informative. DAOs make the mechanism visible, because every vote, delegation, deliberation, and trade is observable at the level of the individual account, but the underlying tension between concentrated control and dispersed information is common to the governance of organizations everywhere.

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Figure 1: Timeline of DAO Governance Events.

This figure illustrates the typical sequence of stages in a DAO governance process, beginning with off-chain forum discussion, followed by proposal creation, voting start, voting end, and on-chain implementation. Arrows indicate the relative timing between consecutive stages.

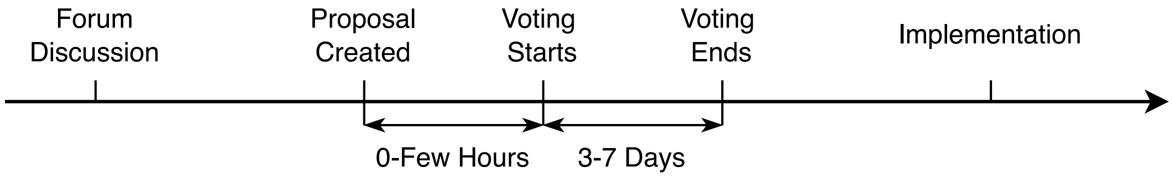


Table 1: Typology of Token Holders

	Low usage intensity ($\delta_h \approx 0$)	High usage intensity (δ_h large)
High financial orientation ($\theta_h \to 1$)	*Financial investors*	*Investor-users*
Low financial orientation ($\theta_h \to 0$)	*Passive holders*	*Protocol users*

Table 2: Summary Statistics

This table presents summary statistics for the key variables that are used in our analyses. For each variable, we report the number of observations, mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Table A.1.

	N	Mean	S.D.	Min	P25	P50	P75	Max
$CAR[-5,5]_i^{created}$	1,593	-0.018	0.269	-1.076	-0.124	-0.022	0.063	5.859
$CAR[-5,5]_i^{end}$	1,658	-0.040	0.216	-1.000	-0.140	-0.029	0.063	1.141
$Participation_i^{Small}$	2,667	0.032	0.045	0.000	0.002	0.011	0.052	0.248
$SmallWin_i$	2,669	0.068	0.251	0.000	0.000	0.000	0.000	1.000
$Weighted_i$	2,830	0.134	0.341	0.000	0.000	0.000	0.000	1.000
$MultipleChoices_i$	2830	0.423	0.494	0.000	0.000	0.000	1.000	1.000
$Quorum_i$	2,830	0.299	0.458	0.000	0.000	0.000	1.000	1.000
$Delegation_i$	2,830	0.264	0.441	0.000	0.000	0.000	1.000	1.000
$Discussion_i$	2,830	0.395	0.489	0.000	0.000	0.000	1.000	1.000
$\Delta Consensus_i$	581	0.136	0.325	-0.987	-0.001	0.038	0.291	0.995
$User_i^{Small}$	2,439	0.572	0.424	0.000	0.022	0.800	0.963	1.000
$ProtocolSecurity_i$	2,830	0.103	0.304	0.000	0.000	0.000	0.000	1.000
$TreasuryExpenditure_i$	2,830	0.573	0.495	0.000	0.000	1.000	1.000	1.000
$UserIncentiveIncrease_i$	2,830	0.272	0.445	0.000	0.000	0.000	1.000	1.000
$Tokenomics_i$	2,830	0.647	0.478	0.000	0.000	1.000	1.000	1.000
$VotingMechanismChange_i$	2,830	0.047	0.211	0.000	0.000	0.000	0.000	1.000

Table 3: Holding Characteristics

This table presents descriptive statistics of variables for small token holders (i.e., eligible voters holding less than 5%) and whales (i.e., eligible voters holding equal or greater than 5%). All variables are defined in Table A.1. The standard errors are clustered at the DAO level. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	Small Token Holder (Holding $<$ 5%)					Whale (Holding $\geq$ 5%)					Small - Whale	
	Mean	Std. Dev.	P10	Median	P90	Mean	Std. Dev.	P10	Median	P90	Diff.	t-stat
$\#\ Holders_i$	6,548.2	8,944.6	1,590.0	3,845.0	15,693.0	2.6	0.9	2.0	3.0	4.0	6,493.2	2.3**
$Turnout_i$	0.0264	0.0286	0.0008	0.0145	0.0766	0.5292	0.3356	0.0000	0.5000	1.0000	-0.5015	-8.6757***
$Delegator_i$	0.0354	0.0212	0.0070	0.0511	0.0538	0.3271	0.1515	0.0000	0.3750	0.5000	-0.2898	-8.8196***
$Delegatee_i$	0.0109	0.0047	0.0040	0.0140	0.0146	0.5811	0.2932	0.0000	0.7500	0.8750	-0.5698	-5.7091***
$Reason_i$	0.0091	0.0409	0.0000	0.0000	0.0079	0.0447	0.1809	0.0000	0.0000	0.0000	-0.0315	-1.1005
HHI_i^{vn}	0.7307	0.3216	0.1554	0.8711	1.0000	0.7909	0.3944	0.0000	1.0000	1.0000	-0.0052	-0.0444
$Timing_i$	0.3486	0.1392	0.1790	0.3467	0.5110	0.5493	0.3083	0.0730	0.5820	0.9119	-0.1864	-2.2477**
$User_i$	0.6795	0.3766	0.0088	0.9028	1.0000	0.5592	0.3874	0.0000	0.5000	1.0000	0.2782	1.7639*

Table 4: The Determinants of Small Token Holder Participation

This table presents estimates of how proposal characteristics, discussion, and user engagement affect small token holder participation. The dependent variable $Participation_{i,t}^{Small}$ is the share of voters and delegators holding less than 5% of circulating governance token supply among all eligible small token holders. Column (1) reports the baseline specification on the full sample. Column (2) restricts the sample to proposals with at least two discussion posts and replaces $Discussion_{i,t}$ with $\Delta Consensus_{i,t}$, the change in discussion consensus. Column (3) restricts the sample to DAOs whose protocols have a deployed Dapp and adds $User_{i,t}^{Small}$, the share of small voters who have previously interacted with the protocol. All variables are defined in Table A.1. The specification includes DAO-category and year fixed effects. The standard errors are clustered at the DAO-category level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	$Participation_{i,t}^{Small}$		
	Full	Discussion $\geq$ 2	Have Dapp
	(1)	(2)	(3)
$Delegation_{i,t}$	0.023***	0.020***	0.020***
	(0.00)	(0.00)	(0.00)
$Discussion_{i,t}$	0.008***		0.005**
	(0.00)		(0.00)
$\Delta Consensus_{i,t}$		0.006*	
		(0.00)	
$User_{i,t}^{Small}$			0.016***
			(0.00)
$MultipleChoices_{i,t}$	0.002	0.001	0.003
	(0.00)	(0.00)	(0.00)
$Weighted_{i,t}$	-0.011***	-0.030***	-0.013***
	(0.00)	(0.00)	(0.00)
$Quorum_{i,t}$	0.024***	0.008**	0.023***
	(0.00)	(0.00)	(0.00)
Category FE	Y	Y	Y
Year FE	Y	Y	Y
Observations	2,665	556	2,286
Adjusted R^2	0.313	0.721	0.332

Table 5: The Determinants of Small Token Holder Delegation

This table presents estimates of how proposal characteristics, discussion, and user engagement affect small token holder delegation. The dependent variable $Delegation_{i,t}^{Small}$ is the share of small token holders who delegate their voting rights for proposal i. Column (1) reports the baseline specification on the full sample. Column (2) restricts the sample to proposals with at least two discussion posts and replaces $Discussion_{i,t}$ with $\Delta Consensus_{i,t}$, the change in discussion consensus. Column (3) restricts the sample to DAOs whose protocols have a deployed Dapp and adds $User_{i,t}^{Small}$, the share of small-token holder voters who have previously interacted with the protocol. All variables are defined in Table A.1. The specification includes DAO-category and year fixed effects. The standard errors are clustered at the DAO-category level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	$Delegation_{i,t}^{Small}$		
	Full	Discussion $\geq$ 2	Have Dapp
	(1)	(2)	(3)
$Discussion_{i,t}$	0.006***		0.007***
	(0.00)		(0.00)
$\Delta Consensus_{i,t}$		0.004*	
		(0.00)	
$User_{i,t}^{Small}$			0.010***
			(0.00)
$MultipleChoices_{i,t}$	0.001	0.001	0.001
	(0.00)	(0.00)	(0.00)
$Weighted_{i,t}$	0.001	-0.000	0.002
	(0.00)	(0.00)	(0.00)
$Quorum_{i,t}$	0.003	0.001	0.005**
	(0.00)	(0.01)	(0.00)
Category FE	Y	Y	Y
Year FE	Y	Y	Y
Observations	682	277	673
Adjusted R^2	0.798	0.652	0.796

Table 6: Participation and Proposal Topics

This table presents estimates of how proposal topic affects small token holder and whale participation. The dependent variables are $Participation_{i,t}^{Small}$ (Column 1), the share of small token holders who participate in proposal i, and $Participation_{i,t}^{Whale}$ (Column 2), the analogous share for whales. The topic indicators $ProtocolSecurity_{i,t}$, $TreasuryExpenditure_{i,t}$, $UserIncentiveIncrease_{i,t}$, $Tokenomics_{i,t}$, and $VotingMechanismChange_{i,t}$ are dummies that equal one if proposal i is directly relevant to the corresponding topic and zero otherwise. All variables are defined in Table A.1. The specification includes DAO-category and year fixed effects. The standard errors are clustered at the DAO-category level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	(1) $Participation_{i,t}^{Small}$	(2) $Participation_{i,t}^{Whale}$
$ProtocolSecurity_{i,t}$	0.004**	0.013
	(0.00)	(0.01)
$TreasuryExpenditure_{i,t}$	0.000	0.013
	(0.00)	(0.01)
$UserIncentiveIncrease_{i,t}$	0.002*	0.003
	(0.00)	(0.01)
$Tokenomics_{i,t}$	0.003	0.009
	(0.00)	(0.01)
$VotingMechanismChange_{i,t}$	-0.004	-0.043
	(0.00)	(0.03)
Token FE	Y	Y
Year FE	Y	Y
Observations	2,657	2,361
Adjusted R^2	0.643	0.721

Table 7: The Determinants of Small Token Holder Victories

This table presents panel logistic Poisson regression with fixed effects estimates of how proposal characteristics, discussion, and user engagement affect the likelihood of a small-token-holder victory. The dependent variable $SmallWin_{i,t}$ is a dummy variable that equals one if proposal i is classified as a small-token-holder win and zero otherwise. Column (1) reports the baseline specification on the full sample. Column (2) restricts the sample to proposals with at least two discussion posts and replaces $Discussion_{i,t}$ with $\Delta Consensus_{i,t}$, the change in discussion consensus. Column (3) restricts the sample to DAOs whose protocols have a deployed Dapp and adds $User^{Small}_{i,t}$, the share of small-token-holder voters who have previously interacted with the protocol. All variables are defined in Table A.1. The specification includes DAO-category and year fixed effects. The standard errors are clustered at the DAO-category level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	$SmallWin_{i,t}$		
	Full	Discussion $\geq$ 2	Have Dapp
	(1)	(2)	(3)
$Delegation_{i,t}$	1.999***		2.220***
	(0.74)		(0.63)
$Discussion_{i,t}$	0.249		0.175
	(0.43)		(0.36)
$\Delta Consensus_{i,t}$		0.098**	
		(0.04)	
$User^{Small}_{i,t}$			1.237***
			(0.40)
$MultipleChoices_{i,t}$	1.037		1.037
	(0.89)		(0.84)
$Weighted_{i,t}$	3.857***	3.603***	3.936***
	(0.45)	(0.00)	(0.45)
$Quorum_{i,t}$	1.076*	-0.100***	1.113**
	(0.65)	(0.00)	(0.56)
Category FE	Y	Y	Y
Year FE	Y	Y	Y
Observations	2,320	173	2,046
Pseudo R^2	0.561	0.213	0.549

Table 8: Staggered DID: Enabling Delegation and Participation

This table presents staggered difference-in-differences estimates of the effect of enabling delegation on token holder participation. The dependent variables are $Participation_{i,t}^{Small}$ (Column 1), the share of small token holders who participate in proposal i, and $Participation_{i,t}^{Whale}$ (Column 2), the analogous share for whales. $Treat_p$ equals one for DAOs that enable delegation. $Post_t$ equals one for proposals occurring after the DAO first enables delegation. The specification includes cohort-DAO and cohort-year fixed effects. All variables are defined in Table A.1. The standard errors are clustered at the DAO level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	(1) $Participation_{i,t}^{Small}$	(2) $Participation_{i,t}^{Whale}$
$Treat_i \times Post_t$	0.015**	0.037
	(0.01)	(0.08)
Cohort-DAO FE	Y	Y
Cohort-Year FE	Y	Y
Observations	1,645	1,515
Adjusted R^2	0.731	0.587

Figure 2: Dynamic Effects of Enabling Delegation on Small Token Holder Participation.

This figure plots the event-study coefficients from the staggered difference-in-differences specification reported in Table 8. The horizontal axis reports event time relative to the DAO's adoption of delegation ($t = 0$). Dots represent point estimates of the dynamic treatment effect on $Participation_{i,t}^{Small}$ and vertical error bars indicate 90% confidence intervals, with standard errors clustered at the DAO level.

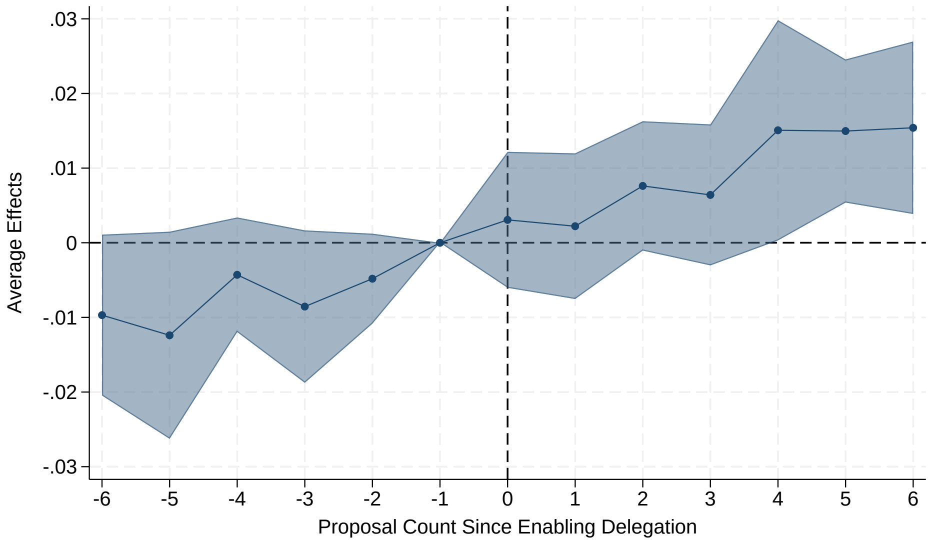


Table 9: Direct and Delegated Participation in Small Holder Victories

This table decomposes small token holder participation into a direct-voting component and a delegated component, for the subsample of proposals in DAOs that enable delegation. $Participation^{Small}$ is the share of small token holders who participate, either by voting directly or by delegating. The direct share is the share who vote directly and the delegated share is the share who delegate, so that the two sum to participation. The delegated fraction is the delegated share divided by participation. The first row reports means across all delegation-enabled proposals; the second row reports means across proposals classified as small token holder victories. All variables are defined in Table A.1.

	Participation *Small*	Direct share	Delegated share	Delegated fraction
All delegation-enabled proposals	0.061	0.026	0.032	0.581
Small shareholder victories	0.066	0.015	0.051	0.772

Table 10: Composition of Voting Power in Small Token Holder Victories

This table reports the average composition of voting power on the winning and losing sides of proposals classified as small token holder victories. For each victory proposal we classify every voter by the share of circulating governance token supply held at the proposal creation block into small token holders ($< 5\%$), small whales (5–10%), and large whales ($\geq 10\%$), and we sum the voting power cast on each side. Entries are the mean percentage of side-level voting power contributed by each group, averaged across victory proposals. The first row uses the vote-count victory definition (vn) used throughout the paper; the second row uses the voting-power victory definition (vp). Voting power cast through delegation is attributed to the delegate, whose own holding determines the group. All variables are defined in Table A.1.

	Winning side			Losing side		
	Small (<5%)	Small whale (5–10%)	Large whale (≥10%)	Small (<5%)	Small whale (5–10%)	Large whale (≥10%)
Vote-count victory (vn)	96.9	0.4	2.7	96.7	2.9	0.4
Voting-power victory (vp)	94.8	3.6	1.6	83.9	12.1	4.1

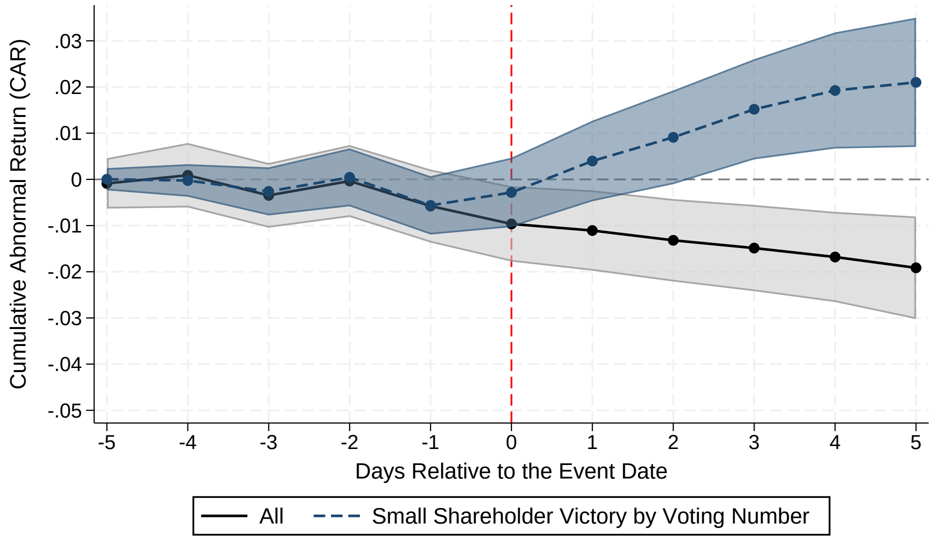
Table 11: Market Reaction to Small Token Holder Victories

This table presents estimates of how small token holder victories affect the market reaction around proposal events. The dependent variables are ${CAR^{Create}}_i$ (Column 1) and ${CAR^{End}}_i$ (Column 2), the cumulative abnormal returns over the $[-5, +5]$ window centered on the proposal creation and voting end dates, respectively, estimated using the CAPM. $SmallWin_{i,t}$ is a dummy variable that equals one if proposal i is classified as a small token holder win and zero otherwise. All variables are defined in Table A.1. The specification includes token and year fixed effects. The standard errors are clustered at the token level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

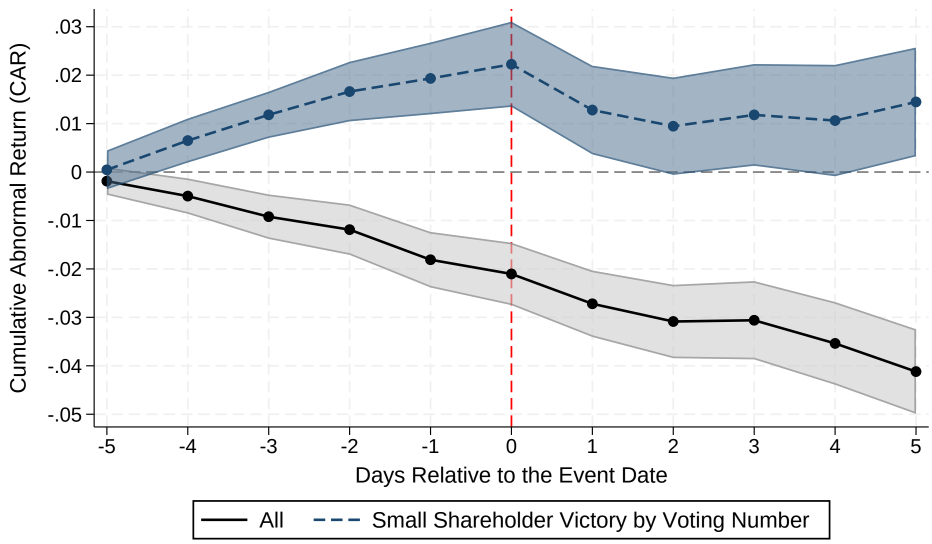
	${CAR^{Create}}_i$	${CAR^{End}}_i$
	(1)	(2)
$SmallWin_{i,t}$	0.046***	0.045***
	(0.01)	(0.01)
Token FE	Y	Y
Year FE	Y	Y
Observations	1,513	1,577
Adjusted R^2	0.222	0.202

Figure 3: Market Reaction to Small Token Holder Victories.

This figure plots the average cumulative abnormal returns of DAO governance tokens around proposal events, separately for proposals classified as small token holder wins and other proposals. Panel (a) reports cumulative abnormal returns centered on proposal creation; Panel (b) reports cumulative abnormal returns centered on voting end. $SmallWin_{i,t}$ equals one if the voting outcome matches the majority choice of small token holders (by vote count) but differs from the majority choice of whales. Abnormal returns are estimated using the CAPM with an estimation window of 250 to 20 trading days prior to the event. Dots represent point estimates and vertical error bars indicate 90% confidence intervals.



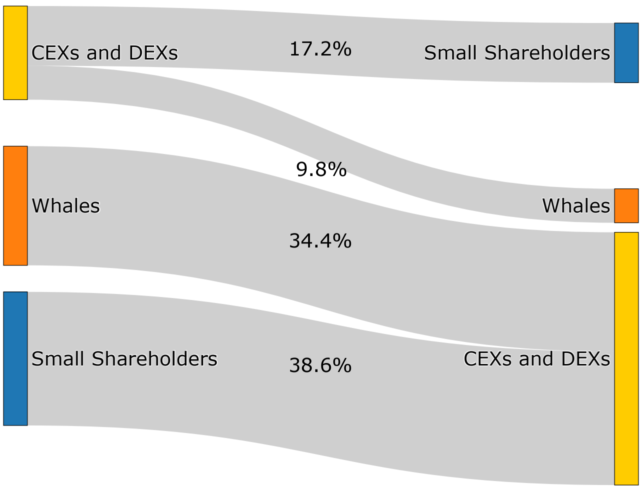
(a) Cumulative Abnormal Returns Around Proposal Creation, $CAR^{Create}{}_i$.



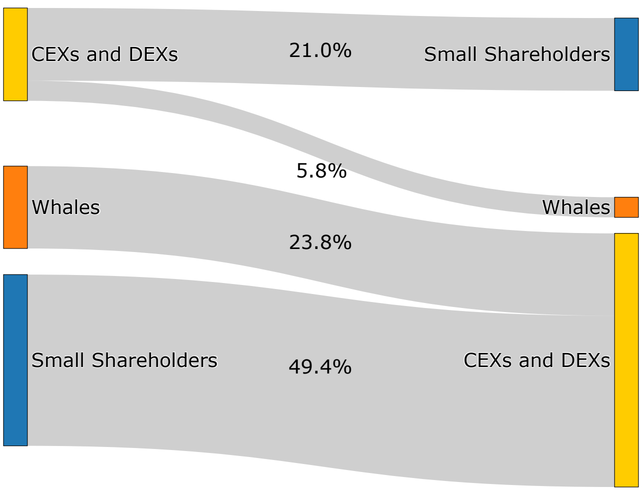
(b) Cumulative Abnormal Returns Around Voting End, $CAR^{End}{}_i$.

Figure 4: Trading Flows of Whales and Small Token Holders Around DAO Governance Events.

This figure presents Sankey diagrams of governance token transactions between whales and small token holders and CEXs and DEXs around proposal events. Panel (a) reports trading flows in the window around proposal creation; Panel (b) reports trading flows in the window around voting end. The width of each band is proportional to the corresponding aggregate trading volume.



(a) Around Proposal Creation.



(b) Around Voting End.

Table 12: Post-Vote Trading by Small Token Holders

This table presents panel logistic Poisson regression with fixed effects estimates of how small-token-holder victories and individual disagreement with the voting outcome relate to post-vote trading by small-token-holder voters. The dependent variables are $Buy_{i,h,t}$ (Columns 1–2), the volume of governance tokens bought by voter h in the days following proposal i, and $Sell_{i,h,t}$ (Columns 3–4), the analogous sell volume. $SmallWin_{i,t}$ is a dummy that equals one if proposal i is classified as a small-token-holder win. $Against\ Outcome_{i,h,t}$ is a dummy that equals one if voter h cast a vote against the final outcome of proposal i. All variables are defined in Table A.1. Columns (1) and (3) absorb voter and year fixed effects and cluster the standard errors at the voter level; Columns (2) and (4) absorb DAO-category and year fixed effects and cluster the standard errors at the DAO-category level. We report t-statistics in parentheses. ***, ** , and * denote significant levels at 1%, 5%, and 10%.

	$Buy_{i,h,t}$		$Sell_{i,h,t}$	
	$SmallWin_{i,t}$	$Against\ Outcome_{i,h,t}$	$SmallWin_{i,t}$	$Against\ Outcome_{i,h,t}$
	(1)	(2)	(3)	(4)
$Small\ Victory^{VN}_{h,t}$	-0.402		-1.321^{***}	
	(0.46)		(0.51)	
$Against\ Outcome_{i,h,t}$		-0.047		0.041
		(0.07)		(0.09)
Voter FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	28,102	190,013	34,406	189,508
Pseudo R^2	0.198	0.088	0.197	0.113

Table 13: Market Reaction to Small Shareholder Victories, by Discussion Quality The sample of proposals with at least two discussion posts is split at the average of the quality measure. Columns (1) and (3) are the below-average (low quality) subsample; columns (2) and (4) the above-average (high quality) subsample. H4 predicts that the $SmallWin_i$ coefficient is larger — and ideally significant only — in the high-quality columns. Proposal category and year fixed effects included; standard errors clustered at the token level. t-statistics in parentheses. ***, **, * denote significance at 1%, 5%, 10%.

	CAR_i^{Create}		CAR_i^{End}	
	Low quality	High quality	Low quality	High quality
	(1)	(2)	(3)	(4)
$SmallWin_i$	0.042	0.044**	0.055***	0.081*
	(0.02)	(0.02)	(0.01)	(0.04)
Category FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	61	91	61	91
Adjusted R^2	0.004	0.045	0.123	0.020

A Appendix

A.1 Variable Definition

Table A.1: Variable Definitions

This table presents definitions for the key variables that are used in our analyses.

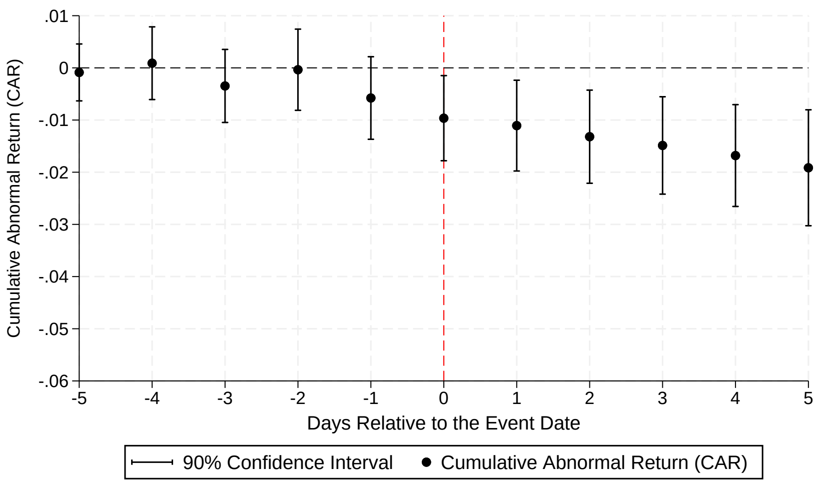
Variable	Description	Data Source
Market Reaction		
$CAR^{Create}{}_i$	Cumulative daily return of DAO governance token of proposal i minus the predicted return, which is estimated by the CAPM. We use an estimation window from 250 to 20 trading days prior to the proposal creation. Each event must have no missing data in the estimation window.	CoinGecko
$CAR^{End}{}_i$	Cumulative daily return of DAO governance token of proposal i minus the predicted return, which is estimated by the CAPM. We use an estimation window from 250 to 20 trading days prior to voting end. Each event must have no missing data in the estimation window.	CoinGecko
DAO Token Holding and Voting		
$\#\ Holders_i$	The number of eligible voters, defined as the number of EOAs holding the governance token of proposal i.	On-chain Data
$Turnout_i$	The percentage of voters relative to the number of DAO governance token holders in proposal i.	Snapshot, On-chain Data
$Delegator_i$	The percentage of delegators relative to the number of voters in proposal i.	On-chain Data
$Delegatee_i$	The percentage of delegatees relative to the number of DAO governance token holders in proposal i.	On-chain Data
$Reason_i$	The percentage of voters who provide a reason when casting their votes, relative to the total number of voters in proposal i.	Snapshot
HHI_i^{vn}	Standardized HHI of vote counts across multiple choices in proposal i.	Snapshot
$Timing_i$	The percentage of the voting period elapsed at the time a voter casts the vote for proposal i.	Snapshot
$User_i$	The percentage of voters who have previously interacted with the smart contract of the DeFi protocol governed by the DAO in proposal i, relative to the total number of voters.	Flipside, On-chain Data
Token Holder Participation		
$User_i^{Small}$	The percentage of small token holders who cast a vote in proposal i and have previously interacted with the smart contract of the DeFi protocol governed by the DAO, relative to the total number of small token holders who cast a vote.	Flipside, On-chain Data

Variable	Description	Data Source
$Participation_i^{Small}$	The percentage of voters and delegators holding less than 5% of circulating governance token supply relative to total eligible voters holding less than 5% of circulating governance token supply in proposal i.	Snapshot, On-chain Data
$Participation_i^{Whale}$	The percentage of voters and delegators holding greater than 5% of circulating governance token supply relative to total eligible voters holding greater than 5% of circulating governance token supply in proposal i.	Snapshot, On-chain Data
$Delegation_i^{Small}$	The percentage of small token holders who delegate their voting rights in proposal i, relative to the total number of small token holders.	Snapshot, On-chain Data
Voting Outcome		
$SmallWin_i$	Dummy variable that equals one if proposal i is classified as a small-token-holder win, defined as when the voting outcome matches the major choice of voters holding less than 5% of circulating supply by vote count but differs from the major choice of voters holding 5% or more of circulating supply by vote count, and zero otherwise.	Snapshot, On-chain Data
Proposal Voting Mechanism		
$Weighted_i$	Dummy variable that equals one if proposal i adopts the weighted voting mechanism and zero otherwise.	Snapshot
$MultipleChoices_i$	Dummy variable that equals one if proposal i adopts the multiple choice voting mechanism and zero otherwise.	Snapshot
$Quorum_i$	Dummy variable that equals one if proposal i requires a quorum and zero otherwise.	Snapshot
$Delegation_i$	Dummy variable that equals one if proposal i enables delegation and zero otherwise.	Snapshot
Proposal Discussion		
$Discussion_i$	Dummy variable that equals one if proposal i has an associated off-chain discussion forum thread and zero otherwise.	Discourse
$\Delta Consensus_i$	Change in the discussion consensus measure for proposal i when the input to the evaluator is expanded from the first half of the replies to the full reply sequence. The consensus measure is scaled from 0 to 1.	Discourse
Post-Vote Trading		
$Buy_{i,h,t}$	Number of governance token purchases made by voter h on day t following proposal i.	On-chain Data
$Sell_{i,h,t}$	Number of governance token sales made by voter h on day t following proposal i.	On-chain Data
$Against\ Outcome_{i,h,t}$	Dummy variable that equals one if voter h cast a vote against the final outcome of proposal i and zero otherwise.	Snapshot

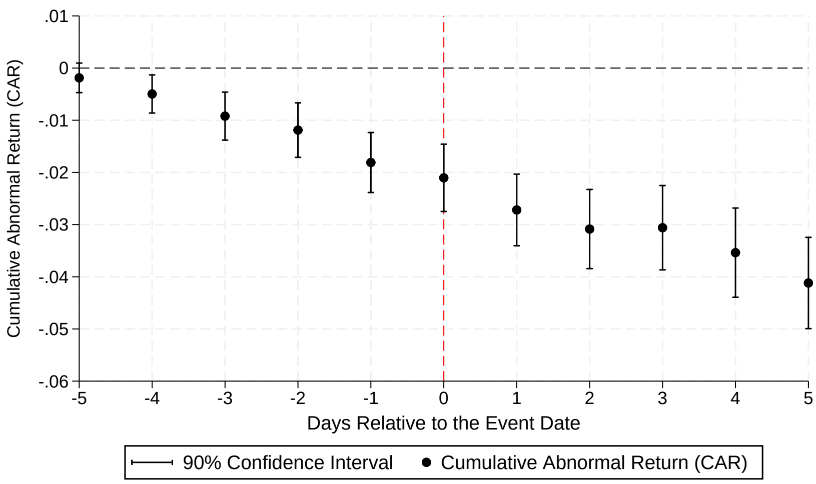
A.2 Market Reaction around Governance Events

Figure A.1: Market Reaction around DAO Governance Events.

This figure plots the average cumulative abnormal returns of DAO governance tokens around proposal events. Panel (a) reports cumulative abnormal returns centered on proposal creation; Panel (b) reports cumulative abnormal returns centered on voting end. Abnormal returns are computed using the CAPM with an estimation window of 250 to 20 trading days prior to the event. The horizontal axis reports trading days relative to the event date ($t = 0$). Dots represent point estimates and vertical error bars indicate 90% confidence intervals.



(a) Cumulative Abnormal Returns Around Proposal Creation, ${CAR^{Create}}_i$



(b) Cumulative Abnormal Returns Around Voting End, ${CAR^{End}}_i$

A.3 LLM Prompt to Evaluate Discussion Consensus

You are an expert in decentralized autonomous organizations (DAOs). Your response should follow this format: '{"result": <true/false>}'. You are given a DAO governance proposal ("Post") and its associated discussion thread ("Discussion"). Your task is to evaluate the **Discussion** according to the following criterion: **Consensus**. The Post and Discussion entries are numbered. If a discussion entry is a reply to another entry, this is indicated in parentheses (e.g., "reply to 5"). User roles such as [moderator], [admin], or [staff] may appear in square brackets. Focus your evaluation strictly on the Discussion, not on the Post itself. Post: {post} (End of Post). Discussion: {discussion} (End of Discussion).

A.4 Theoretical Appendix

A.4.1 Equilibrium Concept and Model Simplification

To derive clean closed-form results, we work with a simplified version of the model in Section 4 that preserves all economically relevant features. The simplification is applied at two levels, to two different parts of the analysis.

For the information and belief-updating block (Lemmas 1–3), we collapse the population to two representative agents: a financial investor (agent W) holding discrete stake α^W and a protocol user (agent U) holding infinitesimal stake $d\alpha^S$, with $\alpha^W \gg d\alpha^S$. We set $\theta_W = 1$, $\delta_W = 0$, $\theta_U = 0$, and $\delta_U > 0$, corresponding to the pure financial investor and pure protocol user cases in Table 1. These are the boundary cases that generate the sharpest predictions, and we adopt them as a simplification; intermediate types attenuate the conflict but do not overturn the comparative statics.

For the participation and outcome results (Lemma 4 and Propositions 2–3), we retain the continuum of small users from Section 4, indexed by usage intensity δ_U with distribution G on $[0, \delta_{\max}]$; agent U then denotes the representative member of this population, while the single financial investor W stands in for the large holders. We write the user agent's objects with subscript U (δ_U, σ_U^2, B_U^S, and so on); these are the theory's individual small-holder objects (δ_h, σ_h^2, $\ldots$), with subscript U read as the representative member h of the continuum. This layer is what makes the participating mass $\Pr(\delta_U > \bar{\delta})$ and the delegated bloc's pooled voting weight well-defined, since these are population objects that a literal two-agent economy could not express. The belief-updating results carry over user by user: every user shares the same updating rule (Lemma 2), and only the realized usage intensity δ_U, and hence the participation cutoff, varies across the population.

We do not collapse the posting side of the forum to a single agent. If one representative user both produced the forum signal and were its only reader, the noise on that signal would have no interpretation, since a single poster cannot average out their own error. We therefore keep the forum aggregate explicit, as in (12). A mass n of users each draw an independent private signal

$$s_h = \pi^U + \varepsilon_h, \qquad \varepsilon_h \sim \mathcal{N}(0, \sigma_h^2), \qquad \varepsilon_h \perp \varepsilon_{h'} \text{ for } h \neq h', \tag{A.1}$$

where, by Assumption 2, the per-user noise $\sigma_h^2 = \sigma^2(\delta_h)$ is set by usage intensity and does not depend on discussion quality. The forum aggregates the posted signals into

$$\bar{s} = \frac{1}{n}\sum_{h=1}^{n} s_h = \pi^U + \frac{1}{n}\sum_{h=1}^{n} \varepsilon_h, \qquad \mathrm{Var}(\bar{s}) = \frac{\bar{\sigma}^2}{n}, \qquad \bar{\sigma}^2 \equiv \frac{1}{n}\sum_{h=1}^{n} \sigma_h^2, \tag{A.2}$$

the average-signal-noise object of (12). The noise on $\bar{s}$ is not any single user's error but the residual that survives averaging across the n posters; by the law of large numbers $\bar{s} \to \pi^U$ as n grows. Agent U in Lemmas 1–3 is a representative member of this posting population, a representative reader of $\bar{s}$ who is also one of its contributors, rather than the sole producer of the signal.

Breadth and quality enter through separate primitives. Breadth n sets the raw precision of the aggregate,

$$\tau_F(n) \equiv \mathrm{Var}(\bar{s})^{-1} = \frac{n}{\bar{\sigma}^2}, \tag{A.3}$$

which is increasing in breadth, since more posters average out more noise, and is the precision of (12). It does not depend on discussion quality, because per-post noise is set by usage intensity. Quality enters at a separate stage, the credibility weighting of Lemma 2, through the verifiability fractions $\kappa_U(\text{quality}_i) > \kappa_W(\text{quality}_i)$ that govern how much of the signal's precision each type can corroborate and act on. Breadth therefore has an informational role, through $\tau_F(n)$, and, in the participation analysis, a coordination role, through the bloc pivotality $R(n)$, while quality enters only through the credibility weights. No primitive depends on both breadth and quality, so the empirical breadth and quality measures map to distinct objects.

The game proceeds in the following sequence:

1. **Nature** draws π^U from $\mathcal{N}(\bar{\pi}^U, \sigma_\pi^2)$ with $\bar{\pi}^U < -\pi^O$.

2. **The posting users** each observe a private signal $s_h = \pi^U + \varepsilon_h$, $\varepsilon_h \sim \mathcal{N}(0, \sigma_h^2)$, and decide whether to post; posts are aggregated into the public forum signal $\bar{s}$ of (A.2), with breadth-driven precision $\tau_F(n)$.

3. **Agent** W observes the forum signal $\bar{s}$ (if posted) and updates beliefs via Bayes' rule, subject to the credibility discount characterized below.

4. **Both agents** simultaneously decide whether to participate in governance and how to vote.

5. The proposal passes if total voting weight in favor of extraction exceeds $\frac{1}{2}$.

We solve for a Perfect Bayesian Equilibrium (PBE) in which strategies are sequentially rational and beliefs are derived from Bayes' rule wherever possible.

A.4.2 Equilibrium Forum Posting

Lemma 1 (Equilibrium posting)**.** *Under verifiable, non-strategic disclosure, the forum signal is the population aggregate $\bar{s}$ of (A.2), with breadth-driven precision $\tau_F(n)$ of (A.3), and $\bar{s} \to \pi^U$ as n grows.*

Proof. Posting is costless, and we treat it as verifiable, non-strategic disclosure: a user who posts reports their realized signal s_h and cannot misreport it, and posting itself is a feature of the forum technology rather than a strategic choice. The forum then aggregates the n posted signals into $\bar{s}$ as in (A.2). By the law of large numbers the idiosyncratic errors average out, $\frac{1}{n}\sum_h \varepsilon_h \to 0$, so $\bar{s} \to \pi^U$ and the precision of $\bar{s}$ is $\tau_F(n)$, increasing in breadth.[2]

The forum bears on the vote only through W. Absent a forum signal W holds its prior, $\mathbb{E}_W[\pi^U] = \bar{\pi}^U_W$, and under condition (11) votes for extraction, so extraction is the no-forum default. Reading the aggregate, W's posterior mean is $\mathbb{E}_W[\pi^U \mid \bar{s}] = \tilde{\lambda}_W \bar{s} + (1-\tilde{\lambda}_W)\bar{\pi}^U_W$, so W switches to the status quo when the pooled evidence is pessimistic enough, $\bar{s} < \bar{s}_W \equiv -[\pi^O + (1-\tilde{\lambda}_W)\bar{\pi}^U_W]/\tilde{\lambda}_W$. The forum thus blocks extraction precisely when $\bar{s}$ reveals value destruction, which is the empirically relevant region of condition (11). ☐

Remark 1 (Posting beliefs rather than raw signals)**.** *We model posts as disclosures of raw usage signals s_h. In practice users may post processed opinions, that is, posterior beliefs about π^U, rather than unfiltered experience. This changes nothing essential. Each user's posterior mean is an affine, invertible transform of their signal, $\mathbb{E}_h[\pi^U \mid s_h] = \lambda_U s_h + (1-\lambda_U)\bar{\pi}^U_U$, so*

[2] We take non-strategic disclosure as a primitive rather than derive it. With many posters who share the status-quo preference, a strategic posting game would have to confront an incentive to withhold optimistic signals: since every user with $\delta_h > 0$ prefers that W block, a user who draws a high signal would prefer not to raise the aggregate toward the point at which W supports extraction. Sustaining full disclosure as an equilibrium would then require additional structure, such as a posting benefit, reputational concerns, or aggregate uncertainty that keeps each post non-pivotal. We instead treat verifiable disclosure as a feature of the forum, the conservative reading for the information channel we study, and do not model the posting game. The same state-independent preference is why verifiability matters at all: were posts unverifiable cheap talk, W would discount a known-biased forum entirely.

a reader who knows the user prior $\bar{\pi}_U^U$ and weight λ_U can recover the implied signal, and the aggregation in (A.2) goes through in the recovered signals. If instead the reader naively averages reported posteriors, the aggregate is pulled toward the users' pessimistic prior $\bar{\pi}_U^U < \bar{\pi}_W^U$; because this makes the forum more pessimistic than a clean signal of π^U, it only widens the credibility wedge in Lemma 2 and reinforces the central result. The raw-signal formulation is therefore the conservative case, and we adopt it for tractability.

A.4.3 Equilibrium Belief Updating

Given Lemma 1, the forum produces the public aggregate $\bar{s} = \pi^U + \frac{1}{n}\sum_h \varepsilon_h$, a noisy signal of the unobservable component with breadth-driven precision $\tau_F(n)$. The two agents extract different information from it because they differ in their ability to verify it. Agent U holds a private signal from direct protocol usage and can validate the forum content against their own experience, so they weight it by λ_U. Agent W holds no private signal about π^U and cannot verify the forum content against direct experience; they therefore apply a smaller credibility weight $\tilde{\lambda}_W \in (0, 1)$. Breadth and quality play distinct roles here, mirroring the main text: breadth n raises the raw informativeness $\tau_F(n)$ of the aggregate and so lifts both weights, while quality governs the differential verifiability that opens the wedge between them. We characterize the key properties of the two weights below.

Lemma 2 (Differential updating)**.** *The equilibrium credibility weights satisfy $\lambda_U > \tilde{\lambda}_W$ for all values of* quality_i*, and both weights are strictly increasing in* quality_i*. Moreover, whenever the forum signal is at least as pessimistic as the financial investor's prior, $\bar{s} \le \bar{\pi}_W^U$, agent W's posterior about π^U is strictly more optimistic than agent U's posterior:*

$$\mathbb{E}_W[\pi^U \mid \bar{s}] > \mathbb{E}_U[\pi^U \mid \bar{s}]. \tag{A.4}$$

Proof. Let σ_π^{-2} denote the prior precision on π^U, common to both agents (their prior means differ by Assumption 3), and let $\tau_F(n) > 0$ be the precision of the forum aggregate $\bar{s}$ from (A.3), increasing in breadth: more posters average out more noise, so a broader forum is more informative about π^U. This raw informativeness is common to both agents; what differs is how much of it each can verify and act on. Each type $t \in \{U, W\}$ treats the forum as carrying the fraction $\kappa_t(\text{quality}_i) \in (0, 1]$ of its precision that it can corroborate, where

$$\kappa_U(\text{quality}_i) > \kappa_W(\text{quality}_i), \qquad \kappa_U' > 0, \quad \kappa_W' > 0, \tag{A.5}$$

for all quality_i: the active user, holding a private usage signal, verifies a larger share than the investor, who holds none, and higher-quality discussion is easier for either type to corroborate.

Each type then updates by Bayes' rule, weighting the forum by the verified precision relative to total precision,

$$\lambda_U = \frac{\kappa_U(\text{quality}_i)\,\tau_F(n)}{\sigma_\pi^{-2} + \kappa_U(\text{quality}_i)\,\tau_F(n)}, \qquad \tilde{\lambda}_W = \frac{\kappa_W(\text{quality}_i)\,\tau_F(n)}{\sigma_\pi^{-2} + \kappa_W(\text{quality}_i)\,\tau_F(n)}. \tag{A.6}$$

The map $x \mapsto x/(\sigma_\pi^{-2} + x)$ is strictly increasing, so $\kappa_U > \kappa_W$ gives $\lambda_U > \tilde{\lambda}_W$ for all quality$_i$, which is Assumption 5(i). Both weights are strictly increasing in quality$_i$, since higher quality raises κ_U and κ_W, and in breadth, since higher n raises τ_F; this is Assumption 5(ii). Breadth raises both weights through the common factor $\tau_F(n)$ and preserves the ordering $\lambda_U > \tilde{\lambda}_W$, while the credibility wedge is created by quality, through the gap $\kappa_U - \kappa_W$. The weights thus depend on quality and breadth through separate primitives, as the channel separation of the main text requires.[3]

For the posterior inequality, combine the credibility gap $\lambda_U > \tilde{\lambda}_W$ with the prior gap $\bar{\pi}^U_W > \bar{\pi}^U_U$ (Assumption 3) and write the posterior difference as a function of $\bar{s}$:

$$\begin{aligned} g(\bar{s}) &\equiv \mathbb{E}_W[\pi^U \mid \bar{s}] - \mathbb{E}_U[\pi^U \mid \bar{s}] \\ &= \tilde{\lambda}_W \bar{s} + (1-\tilde{\lambda}_W)\bar{\pi}^U_W - \lambda_U \bar{s} - (1-\lambda_U)\bar{\pi}^U_U \\ &= (\tilde{\lambda}_W - \lambda_U)\,\bar{s} + (1-\tilde{\lambda}_W)\bar{\pi}^U_W - (1-\lambda_U)\bar{\pi}^U_U. \end{aligned} \tag{A.7}$$

The function g is affine in $\bar{s}$ with slope $\tilde{\lambda}_W - \lambda_U < 0$, so it is strictly decreasing. It is therefore positive if and only if $\bar{s} < \bar{s}^*$, where the threshold is

$$\bar{s}^* = \frac{(1-\tilde{\lambda}_W)\bar{\pi}^U_W - (1-\lambda_U)\bar{\pi}^U_U}{\lambda_U - \tilde{\lambda}_W}. \tag{A.8}$$

The threshold exceeds the investor's prior. Indeed,

$$\begin{aligned} \bar{s}^* - \bar{\pi}^U_W &= \frac{(1-\tilde{\lambda}_W)\bar{\pi}^U_W - (1-\lambda_U)\bar{\pi}^U_U - (\lambda_U - \tilde{\lambda}_W)\bar{\pi}^U_W}{\lambda_U - \tilde{\lambda}_W} \\ &= \frac{(1-\lambda_U)\big(\bar{\pi}^U_W - \bar{\pi}^U_U\big)}{\lambda_U - \tilde{\lambda}_W} > 0, \end{aligned} \tag{A.9}$$

since $\lambda_U < 1$, $\lambda_U > \tilde{\lambda}_W$, and $\bar{\pi}^U_W > \bar{\pi}^U_U$ by Assumption 3. Hence any $\bar{s} \le \bar{\pi}^U_W < \bar{s}^*$ gives $g(\bar{s}) > 0$, which is the posterior inequality (A.4). □

The condition $\bar{s} \le \bar{\pi}^U_W$ is the empirically relevant region. When the forum aggregates genuine evidence of value destruction, the public signal $\bar{s}$ comes in below the investor's optimistic prior, and the result then holds. In this region the signal term $(\tilde{\lambda}_W - \lambda_U)\bar{s}$ in (A.7) is positive and

[3] Assumption 5 is, in the reduced model, a consequence of the single primitive $\kappa_U > \kappa_W$ with $\kappa_U', \kappa_W' > 0$: this lemma derives both the credibility ordering and the monotonicity in quality from it. Setting $\kappa_U \equiv 1$ recovers the special case in which the user verifies the forum fully.

reinforces the prior gap: because agent U places more weight on the pessimistic signal than agent W does, the same forum report leaves agent W more optimistic about π^U. The ranking in fact persists on the larger region $\bar{s} < \bar{s}^*$, and since $\bar{s}^* > \bar{\pi}^U_W$ by (A.9), the condition $\bar{s} \leq \bar{\pi}^U_W$ is a clean sufficient condition rather than the sharp threshold; the ranking reverses only for $\bar{s} > \bar{s}^*$, when the forum reports news markedly milder than the investor already believed. That case is not relevant for the governance failure we study, which concerns proposals where the forum reveals that π^U is strongly negative.

A.4.4 Equilibrium Voting and Participation

Lemma 3 (Equilibrium voting strategies)**.** *In any PBE: agent W votes for extraction if and only if $\pi^O + \mathbb{E}_W[\pi^U \mid \bar{s}] > 0$; agent U votes against extraction for all $\delta_U > 0$.*

Proof. For agent W: substituting $\theta_W = 1$ and $\delta_W = 0$ into condition (6) gives $\alpha^W(\pi^O + \mathbb{E}_W[\pi^U \mid \bar{s}]) > 0$, which simplifies to the stated condition since $\alpha^W > 0$. For agent U: substituting $\theta_U = 0$ into condition (6) gives $0 > -\delta_U$, which holds strictly for all $\delta_U > 0$ regardless of agent U's posterior about π^U. □

Lemma 4 (Equilibrium participation)**.** *In the absence of governance infrastructure, agent W participates and agent U does not. With governance infrastructure:*

(i) Agent U participates through delegation if and only if $B^{\text{del}}_U > c_{\text{del}}$, where B^{del}_U is given by (15) and $c_{\text{del}} \ll c$; and

(ii) There exists a threshold $\bar{\delta}(n, \text{quality}_i, c_{\text{del}})$ such that users with $\delta_U > \bar{\delta}$ participate and those with $\delta_U \leq \bar{\delta}$ do not, where $\bar{\delta}$ is strictly decreasing in n and quality_i and strictly increasing in c_{del}.

Proof. Without infrastructure: agent W has $\Pr(\text{pivotal}_W) > 0$ and $\alpha^W|\pi^O + \mathbb{E}_W[\pi^U]| > c$ for sufficiently large α^W, so participation is individually rational. Agent U has $\Pr(\text{pivotal}_U) \to 0$ as $d\alpha^S \to 0$, so $B^S_U \to 0 < c$ and agent U does not participate.

With delegation: from condition (15) and $\theta_U = 0$, the participation condition reduces to $B^{\text{del}}_U = R(n)\,\Lambda(\text{quality}_i)\,\delta_U > c_{\text{del}}$, where $R(n) \equiv \mathbb{E}[\mathbf{1}[\text{bloc pivotal}]]$. Rearranging gives the threshold $\bar{\delta} = c_{\text{del}}/(R(n)\,\Lambda(\text{quality}_i))$. This threshold is increasing in c_{del} by construction. Higher forum quality raises $\Lambda(\text{quality}_i)$, established in (A.11) below, where its monotonicity is derived from Lemma 2 rather than assumed, and so reduces $\bar{\delta}$. Higher forum breadth n raises the bloc pivotality $R(n)$, raising B^{del}_U and reducing $\bar{\delta}$; breadth also sharpens the aggregate through $\tau_F(n)$, which raises Λ as well, so both effects push $\bar{\delta}$ in the same direction. For the comparative statics we attribute the participation effect of breadth to the coordination channel $R(n)$ and treat its informational effect as reinforcing. This is conservative and leaves quality, which enters Λ only through κ_U, as the informational margin in the empirical design. Part (ii) follows. $\square$

The information-value factor. We give the information-value factor Λ in (15) an explicit definition, so that its dependence on quality is derived from Lemma 2 rather than posited. A user contemplating delegation values carrying their usage-aligned preference into a pivotal outcome in proportion to how confident they are, after reading the forum, that the proposal is value-destroying. We take this confidence to be the precision of their post-forum posterior about π^U and set Λ equal to it:

$$\Lambda(\text{quality}_i, n) \equiv \text{Var}_U[\pi^U \mid \bar{s}]^{-1} = \sigma_\pi^{-2} + \kappa_U(\text{quality}_i)\, \tau_F(n), \tag{A.10}$$

the post-forum posterior precision of agent U, by the Gaussian updating of Lemma 2. Any increasing affine transformation would give the same comparative statics; we use the precision itself. It is increasing in quality through κ_U and in breadth through τ_F,

$$\frac{\partial \Lambda}{\partial\, \text{quality}_i} = \kappa_U'(\text{quality}_i)\, \tau_F(n) > 0, \qquad \frac{\partial \Lambda}{\partial n} = \kappa_U(\text{quality}_i)\, \tau_F'(n) > 0, \tag{A.11}$$

so Λ rises in quality through the same primitive ($\kappa_U' > 0$) that drives λ_U. The step "higher λ_U, sharper posterior, higher Λ" used in the participation proofs is therefore exact rather than heuristic, since λ_U and Λ are both increasing in $\kappa_U(\text{quality}_i)\, \tau_F(n)$.

A.4.5 Proofs of Main Propositions

Proof of Proposition 1. By Lemma 4, agent U does not participate without infrastructure. By Lemma 3, agent W votes for extraction whenever $\pi^O + \mathbb{E}_W[\pi^U] > 0$. Without a forum signal, $\mathbb{E}_W[\pi^U] = \bar{\pi}_W^U$, so this condition becomes $\bar{\pi}_W^U > -\pi^O$, which is condition (11) imposed in the proposition. Under that condition, and since agent W is the only voter, extraction passes despite $\Delta V = \pi^O + \pi^U < 0$. This constitutes a PBE. $\square$

Proof of Proposition 2. Part (i): Coordination channel. By Lemma 4(ii), $\bar{\delta}$ is decreasing in n because the bloc pivotality $R(n)$ is increasing in n: broader forum participation coordinates a larger delegated bloc, which is more likely to be decisive. As $\bar{\delta}$ falls, the set $\{\delta_U > \bar{\delta}\}$ of participating users expands strictly. The effect is positive for any $c_{\text{del}} \leq c$.

Part (ii): Information aggregation channel. Higher quality_i increases λ_U (Lemma 2), sharpening agent U's posterior. For users with δ_U close to $\bar{\delta}$, the improved posterior reduces uncertainty about whether participation is worthwhile, strictly increasing the expected benefit B_U^{del} and reducing $\bar{\delta}$. The set of participating users expands strictly.

Part (iii): Delegation effect. Replacing c with $c_{\text{del}} \ll c$ in Lemma 4 strictly reduces $\bar{\delta}$. The set of participating users expands discontinuously at the moment delegation is enabled.

Complementarity. We establish the complementarity at the level of the participation benefit, which is the object Topkis monotone comparative statics applies to, rather than at the level of the threshold, whose inverse relationship to the benefit makes its cross-partial uninformative. For a user of type δ_U, the benefit from participating through the bloc takes the multiplicative form

$$B_U^{\text{del}}(n, \text{quality}_i; \delta_U) = R(n) \cdot \Lambda(\text{quality}_i, n) \cdot \delta_U, \tag{A.12}$$

as in (15) with $\theta_U = 0$, where $R(n) \equiv \mathbb{E}[\mathbf{1}[\text{bloc pivotal}]]$ is the bloc pivotality, increasing in forum breadth, and Λ is the information-value factor of (A.10), increasing in both quality and breadth. Both R and Λ depend on n, so the cross-partial carries two terms,

$$\frac{\partial^2 B_U^{\text{del}}}{\partial n\, \partial \text{quality}_i} = \Big[R'(n) \frac{\partial \Lambda}{\partial \text{quality}_i} + R(n) \frac{\partial^2 \Lambda}{\partial n\, \partial \text{quality}_i} \Big] \delta_U = \kappa_U'(\text{quality}_i) \big[R'(n)\, \tau_F(n) + R(n)\, \tau_F'(n) \big] \delta_U > 0, \tag{A.13}$$

using $\partial\Lambda/\partial\text{quality}_i = \kappa_U' \tau_F$ and $\partial^2\Lambda/\partial n\,\partial\text{quality}_i = \kappa_U'\tau_F'$ from (A.11). Both terms are positive, since $R, R', \tau_F, \tau_F', \kappa_U', \delta_U > 0$, so the benefit is supermodular in $(n, \text{quality}_i)$: the marginal increase in the participation benefit from greater forum breadth is strictly larger when forum quality is higher, and conversely. Equation (A.13) is the breadth×quality cross-partial; the delegation×quality complementarity that the empirical interaction tests (delegation interacted with discussion quality) is the discrete-margin statement established next, via the participating measure $\Pr(\delta_U > \bar{\delta})$.

This delivers the two complementarity claims of Proposition 2. First, the participation gain from enabling delegation is the measure of users brought into participation. Without delegation, atomistic users are non-pivotal and do not participate at any forum breadth or quality. With delegation, the set of participants is $\{\delta_U > \bar{\delta}(n, \text{quality}_i)\}$, with $\bar{\delta} = c_{\text{del}}/(R(n)\Lambda(\text{quality}_i))$, whose measure $\Pr(\delta_U > \bar{\delta})$ is increasing in quality_i because $\bar{\delta}$ is decreasing in quality_i. The gain from delegation is therefore strictly increasing in forum quality. Second, the participation gain from greater forum breadth is zero without delegation (atomistic users never participate) and strictly positive with delegation, hence strictly larger when delegation is enabled. Both claims follow from $\bar{\delta}$ being strictly decreasing in n and in quality_i separately, which the benefit's supermodularity in (A.13) underlies, together with the discrete delegation margin.[4] □

Proof of Proposition 3. The probability of a user-preferred outcome equals the probability that the user bloc's pooled voting weight exceeds agent W's weight on the extraction side. By

[4] We do not claim the participation rate $\Pr(\delta_U > \bar{\delta})$ is itself supermodular in $(n, \text{quality}_i)$; that depends on the curvature of the type distribution G and is neither needed nor asserted.

Lemma 3, all participating users vote against extraction. By Lemma 4, the mass of participating users is $\Pr(\delta_U > \bar{\delta})$, which is strictly decreasing in $\bar{\delta}$. Since $\bar{\delta}$ is strictly decreasing in n and quality_i and strictly increasing in c_{del} (Lemma 4(ii)), the user bloc's aggregate voting weight, and therefore the probability of a user-preferred outcome, is strictly increasing in n and quality_i and strictly decreasing in c_{del}.

The composition result follows because the newly participating users at the margin have $\delta_U = \bar{\delta}$, which falls as n or quality_i rises or c_{del} falls. As $\bar{\delta}$ falls, the marginal participant has progressively higher usage intensity, and since signal precision is increasing in δ_U (Assumption 2), the composition of the voter pool shifts toward better-informed agents. □

Proof of Proposition 4. Throughout, fix the value-destroying proposal class of Assumption 1, and read that assumption as a restriction on the realization of π^U, not only its mean. Let $\mathcal{D} \equiv \{\pi^U : \pi^U < -\pi^O\}$ be the event that the proposal is value-destroying ex post, so $\Delta V = \pi^O + \pi^U < 0$ on $\mathcal{D}$. The empirical design conditions on contested proposals that the forum and the ex-post outcome identify as value-destroying, so the relevant population is the law of π^U truncated to $\mathcal{D}$, and all expectations below are taken under that truncated law.[5]

Positive CAR. When informed users block extraction, the status quo is preserved and the market prices the avoided change in token value. On $\mathcal{D}$ we have $\pi^U < -\pi^O < 0$, hence $|\pi^U| > \pi^O$ and the avoided loss $|\pi^U| - \pi^O$ is strictly positive for every realization in $\mathcal{D}$. Since the user-victory event $\mathcal{I}_{\text{win}} \subseteq \mathcal{D}$, the conditional expectation of a strictly positive integrand is strictly positive:

$$\text{CAR} = \mathbb{E}\big[|\pi^U| - \pi^O \mid \mathcal{I}_{\text{win}}\big] > 0. \tag{A.14}$$

The positivity is established on the truncated class, on which $|\pi^U| - \pi^O > 0$ pointwise; it does not require that a user victory select more value-destroying proposals within $\mathcal{D}$, and it holds for the avoided loss itself.

Increasing in forum quality. The magnitude of the reaction depends on how large the long-horizon market participants assess the avoided loss to be. We model these participants as a third group, distinct from the voters W and U: they are rational, do not observe π^U directly, and read the forum signal $\bar{s}$. They form a posterior on π^U with mean

$$\mu_M(\bar{s}) = \lambda_M(\text{quality}_i)\, \bar{s} + \big(1 - \lambda_M(\text{quality}_i)\big)\, \bar{\pi}^U$$

[5]Equivalently, read Assumption 1 as $\text{supp}(\pi^U) \subseteq \mathcal{D}$ for the relevant class, rather than as a restriction on the unconditional Gaussian mean alone. This is the reading under which the conditional-event statements of the proposition are well posed: the pointwise positivity and the linearization below both rest on $\pi^U < 0$, which holds on $\mathcal{D}$ but not on the untruncated Gaussian.

and dispersion σ_M, where $\lambda_M(\text{quality}_i) \in (0,1)$ is a market credibility weight, strictly increasing in quality, for the same verification reason as in Assumption 5.[6]

The market prices the avoided loss conditional on the value-destroying class, $\pi^U \in \mathcal{D} = \{\pi^U < -\pi^O\}$. For a normal posterior, truncation to $\pi^U < -\pi^O$ shifts the mean by σ_M times the inverse Mills ratio of the standard normal,

$$\mathbb{E}[\pi^U \mid \bar{s}, \mathcal{D}] = \mu_M(\bar{s}) - \sigma_M\, m(\beta), \qquad \beta \equiv \frac{-\pi^O - \mu_M(\bar{s})}{\sigma_M}, \qquad m(\beta) \equiv \frac{\phi(\beta)}{\Phi(\beta)}, \tag{A.15}$$

where ϕ and Φ are the standard normal density and distribution and $m(\cdot)$ is the hazard rate of the standard normal. On $\mathcal{D}$ the avoided loss is $|\pi^U| - \pi^O = -\pi^U - \pi^O$, so

$$\text{CAR} = \mathbb{E}\big[-\pi^U - \pi^O \mid \mathcal{I}_{\text{win}}\big] = \sigma_M\, m(\beta) - \mu_M(\bar{s}) - \pi^O. \tag{A.16}$$

Positivity holds exactly: a left-truncated normal mean lies strictly below the truncation point, $\mathbb{E}[\pi^U \mid \bar{s}, \mathcal{D}] < -\pi^O$, so $\text{CAR} = -\mathbb{E}[\pi^U \mid \bar{s}, \mathcal{D}] - \pi^O > 0$, consistent with (A.14).

For the comparative static, the truncated mean is strictly increasing in the untruncated mean, $\partial\, \mathbb{E}[\pi^U \mid \bar{s}, \mathcal{D}]/\partial\, \mu_M = \text{Var}[\pi^U \mid \bar{s}, \mathcal{D}]/\sigma_M^2 \in (0,1)$, the ratio of the truncated to the untruncated variance.[7] In the empirically relevant region the forum is more pessimistic than the market's prior, $\bar{s} < \bar{\pi}^U$, so $\partial\, \mu_M/\partial\, \lambda_M = \bar{s} - \bar{\pi}^U < 0$: more credible discussion pulls the market's mean down. Hence

$$\frac{\partial\, \text{CAR}}{\partial\, \lambda_M} = -\frac{\partial\, \mathbb{E}[\pi^U \mid \bar{s}, \mathcal{D}]}{\partial\, \mu_M} \cdot \frac{\partial\, \mu_M}{\partial\, \lambda_M} > 0,$$

the product of a term in $(-1, 0)$ and a negative term. Since λ_M is strictly increasing in quality_i, $\partial\, \text{CAR}/\partial\, \text{quality}_i > 0$. This presumes the forum signal is not already fully impounded in the price before the vote, so that a user victory resolves residual uncertainty about whether extraction is blocked; this holds whenever the governance outcome is uncertain ex ante. □

[6] This is an explicit assumption on the market, not a consequence of the voting model, since the long-horizon participants are outside the two-type model. We state it as a primitive: higher-quality discussions are easier to corroborate without a private signal, hence more credible to any uninformed observer. We take the posterior dispersion σ_M as a fixed parameter, so that quality acts on the mean the market reads from the forum.

[7] This is the standard truncated-normal identity: shifting the location of a normal shifts its truncated mean by the truncated-variance share, which lies strictly between zero and one.